

A FISCAL THEORY OF PERSISTENT INFLATION*

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We develop a new class of general-equilibrium models with partially unfunded debt to propose a fiscal theory of persistent inflation. In response to business cycle shocks, the monetary authority controls inflation and the fiscal authority stabilizes debt. However, the central bank accommodates unfunded fiscal shocks, causing persistent movements in inflation, output, and real interest rates. In an estimated quantitative model, fiscal inflation accounts for the bulk of inflation dynamics. In the aftermath of the pandemic, unfunded fiscal shocks sustain the recovery but also cause a persistent increase in inflation. The model is able to predict the inflationary effects of the American Rescue Plan Act fiscal stimulus out of sample and with real-time data. *JEL codes:* E50, E62, E30.

Persistent high inflation is always and everywhere a fiscal phenomenon.

—Thomas J. Sargent (2013)

I. INTRODUCTION

We build a novel class of general-equilibrium models with partially unfunded debt to propose a fiscal theory of persistent inflation. In response to business cycle shocks and funded fiscal shocks, the monetary authority controls inflation and the fiscal authority stabilizes debt. However, the central bank

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accommodates unfunded fiscal shocks, causing persistent movements in inflation and real interest rates.

We first illustrate the distinction between funded and unfunded fiscal shocks in the context of a Fisherian model. Funded and unfunded shocks coexist in the model. The difference between the two types of shocks is that funded fiscal shocks are backed by future fiscal adjustments, and unfunded fiscal shocks are not. As a result, funded fiscal shocks are irrelevant for inflation, while unfunded fiscal shocks lead to an increase in inflation accommodated by the central bank. We consider the effects of unfunded fiscal shocks in production economies. Absent nominal rigidities, unfunded fiscal shocks cause large and temporary jumps in inflation and do not have real effects, as in the Fisherian model. Allowing the central bank to partially respond to fiscal inflation delivers a more persistent inflation response, while introducing a maturity structure tempers the size of the initial jump. However, these extensions leave real activity and real interest rates unaffected and create a counterfactually tight link between inflation persistence and the monetary policy rule. Instead, with nominal rigidities, unfunded fiscal shocks have real effects and cause persistent movements in inflation and real interest rates, which leads us to a fiscal theory of persistent inflation.

We augment a quantitative New Keynesian (NK) model with unfunded fiscal shocks to assess their importance for U.S. inflation dynamics. The model features all the ingredients that have been proven successful in matching U.S. business cycle dynamics, including a large set of business cycle shocks. With regard to these shocks, monetary policy satisfies the Taylor principle and the fiscal authority is in control of debt stabilization. Thus, in this respect, the model behaves like its counterparts extensively studied in the literature. However, the model also features unfunded fiscal shocks. We model these as shocks to transfers that are not backed by future fiscal adjustments, implying that a share of the overall government debt is unfunded. The central bank accommodates the increase in inflation necessary to stabilize the unfunded amount of debt. As in the textbook version of the NK model, these shocks trigger persistent movements in inflation, output, and real interest rates.

The model is endowed with a rich set of shocks, including a persistent shifter to the New Keynesian Phillips curve (NKPC), which is meant to capture autonomous factors, such as globalization and demographic changes, which can affect inflation in the

long run. The model also features hand-to-mouth agents to allow for the possibility that funded fiscal shocks can influence macroeconomic dynamics. Thus, it is an empirical question whether unfunded fiscal shocks play an important role in explaining the data. We show that they do.

A persistent and partially unfunded increase in transfers in the mid-1960s, related to the introduction of the Great Society initiatives, accounts for the persistent rise in inflation during the Great Inflation. Symmetrically, the end of the Great Inflation is explained by a sharp revision in the amount of inflation that the Federal Reserve was going to tolerate to stabilize the portion of unfunded debt. In this respect, the aggressive increase in interest rates implemented by the Federal Reserve under Chair Paul Volcker can be interpreted as a strong signal of this policy change. After that, fiscal inflation did not completely disappear. The amount of unfunded spending increased sluggishly starting from the 1990s, and then it accelerated after the Great Recession. However, during these years fiscal inflation was to some extent beneficial because it counteracted a deflationary bias due to nonpolicy shocks, arguably related to demographics and international trade.

Unfunded spending experienced a new acceleration during the COVID-19 pandemic. A large part of the economic rebound at the end of 2020 is attributed to the \$2.2 trillion fiscal package introduced in March of that year to combat the consequences of the pandemic crisis, and that package we find to be partially unfunded. Without these unfunded shocks, the economy would have experienced deflation. Interestingly, a shift in the portion of unfunded spending is observed in the third and fourth quarters of 2020, after the Federal Reserve announced its new operating framework that contemplates the possibility of letting inflation overshoot its 2% target after the pandemic recession.¹ This new monetary policy strategy is reflected in the market expected path of the federal funds rate, which we observe in the estimation. Finally, with the economy already on a recovery path, the 2021 \$1.9 trillion American Rescue Plan Act (ARPA) fiscal stimulus

1. On August 27, 2020, Federal Reserve Chair Jerome Powell (2020) announced the new framework for the first time at the Jackson Hole Economic Symposium as follows: "Following periods when inflation has been running below 2 percent, appropriate monetary policy will likely aim to achieve inflation moderately above 2 percent for some time."

determines a further increase in the amount of unfunded spending, exacerbating the postpandemic inflation increase.

As an important validation, we show that the model correctly predicts the inflationary effects of the ARPA even out of sample and when using real-time data as available in the second quarter of 2021. We proceed in two steps. First, we produce a projection for inflation using only estimates and data up to the end of 2020. This projection reflects the effects of the the Coronavirus Aid, Relief, and Economic Security Act (CARES Act). Under this scenario, the model predicts a modest inflation overshoot for the next four years, with a peak of 3.7%, and then a gradual return to the 2% target. We consider the effects of the ARPA. We assign part of the increase in spending in the first quarter of 2021 to unfunded transfers based on the historical evidence. We construct a conditional forecast and show that the model can replicate very closely the inflation figures for 2021. The model now predicts a substantially more robust overshoot of inflation, with a peak of 6.3% instead of 3.7%, plus a very slow return to the 2% target.

Based on these results, we conclude that unfunded spending has played an important role in accounting for inflation dynamics, both historically and after the pandemic. From this point of view, the postpandemic situation is not necessarily different from the historical experience of the United States. However, two qualifications have to be made. First, in the postmillennial period, fiscal inflation has been counteracting a deflationary bias. The risk of persistent high inflation depends in part on whether this deflationary bias will persist. If it were to disappear, perhaps because of a change in the degree of integration of the world economy, the postmillennial high level of fiscal inflation could become a problem. Second, as of 2023, spending is at a historical maximum. This implies that even small changes in the share of unfunded debt can lead to large swings in inflation.

In the last part of the article, we conduct a counterfactual simulation to highlight the key mechanisms that make unfunded fiscal shocks quantitatively important. We trim down the quantitative baseline model to only retain nominal rigidities as in a textbook NK model. We show that when this bare-bones NK model is fed with the same unfunded fiscal shocks identified in the data, it delivers a very similar path for inflation, suggesting that the additional features of our baseline model do not drive the results with respect to the empirical relevance of fiscal inflation. These additional features are included to preserve the propagation

mechanism of the other shocks and to avoid ruling out a priori other explanations for movements in inflation.

From the methodological standpoint, this article develops a new class of models in which policy makers are allowed to react differently to different shocks. The focus of this study is on the effects of unfunded fiscal shocks. However, this new class of models can be used to study other forms of heterogeneity in policy responses. The article builds on [Bianchi and Melosi \(2019\)](#), who introduced the concept of shock-specific rules as a way to resolve a conflict between the monetary and fiscal authorities in the presence of a high fiscal burden that the fiscal authority is reluctant or unable to stabilize. In that work, we apply a shock-specific rule to study the macroeconomic effects of introducing an emergency budget to mitigate a large recession. In this article, we extend the notion of shock-specific rules to solve general-equilibrium models in which monetary and fiscal authorities adopt state-dependent targets. This delivers a fiscal theory of persistent inflation that is always at work, not just in response to exceptional events.

This article is connected to the vast literature on monetary–fiscal policy interaction (among many others, [Sargent and Wallace 1981](#); [Leeper 1991](#); [Sims 1994](#); [Woodford 1994, 1995, 2001](#); [Cochrane 1998, 2001](#); [Schmitt-Grohé and Uribe 2000, 2002](#); [Benhabib, Schmitt-Grohé and Uribe 2002](#); [Bassetto 2002](#); [Reis 2017](#); and [Bassetto and Sargent 2020](#)). [Barro \(1974\)](#) shows that an alternative way to generate non-Ricardian effects is if agents erroneously regard bonds as net wealth. [Aiyagari and Gertler \(1985\)](#) study the implications of fiscal backing of government bonds for the propagation of shocks. They find that for debt to be irrelevant, the model needs to feature a considerable degree of accommodation with respect to the monetary authority. [Leeper and Zhou \(2021\)](#) find that inflation plays an important role in the optimal marginal financing of fiscal needs in models similar to the one used in our empirical analysis. [Hall and Sargent \(2011\)](#) show that historically most U.S. debt stabilization has been achieved through a combination of growth, revaluation effects, and low real interest rates. [Bianchi and Ilut \(2017\)](#) estimate a model with regime changes in the monetary/fiscal policy mix and link the high inflation of the 1960s and 1970s to a fiscally led regime. [Bianchi and Melosi \(2017, 2022\)](#) argue that the possibility of a return to such a regime can explain the lack of deflation in the aftermath of the Great Recession and the postpandemic inflation. In a recent opinion piece, [Barro \(2022\)](#) reaches a conclusion similar to the one proposed here with respect to the

postpandemic inflation: a realignment in prices is necessary to stabilize the large pandemic increase in the fiscal burden.

With respect to models with regime changes in the monetary-fiscal policy mix, we move in a new and different direction. Monetary-led and fiscally led rules coexist in our model, and the policy coordination is shock-specific. Shocks to unfunded transfers are dealt with by fiscally led policies. With respect to all other shocks, the monetary authority controls inflation and the fiscal authority is responsible for debt stabilization. As a result, this new modeling approach delivers low-frequency movements in inflation linked to unfunded fiscal shocks, while at the same time preserving the typical propagation of the business cycle shocks used in NK models. In this regard, our work is related to [Cochrane \(2022\)](#) who considers a class of NK models in which pre-existing debt is backed by future surpluses, but in response to unexpected movements in inflation, policymakers are not committed to make fiscal adjustments. The key difference is that [Cochrane \(2022\)](#) assumes that the policy mix is always fiscally led, while in our setting a monetary-led and a fiscally led policy mix coexist at the same time. The two papers share the goal of integrating elements of the fiscal theory of the price level in an NK framework.

II. PERSISTENT FISCAL INFLATION

In this section, we introduce a new class of models in which a monetary-led and a fiscally led policy mix coexist. The propagation of shocks changes depending on the shock-specific policy response. This allows us to introduce unfunded fiscal shocks in an otherwise standard model. We illustrate the logic of this new class of models with shock-specific rules in the context of a simple Fisherian model ([Leeper 1991](#); [Sims 1994, 2016](#)). Our focus is on fiscal inflation, but the method can be applied in other settings where a researcher is interested in modeling shock-specific policy responses. We move to a production economy and discuss the consequences of introducing nominal rigidities. We show that in a model with nominal rigidities, unfunded fiscal shocks lead to persistent movements in inflation and real effects.

II.A. Funded and Unfunded Fiscal Shocks

1. *Stylized Model.* The economy is populated by a continuum of infinitely many households and a government. The

representative household has concave and twice continuously differentiable preferences over nonstorable consumption goods and is endowed in each period with a constant quantity Y of these goods. The government issues one-period debt B_t to households that can trade them for consumption goods. The representative household chooses consumption and government bonds so as to maximize:

$$\max \sum_{t=0}^{\infty} \beta^t U(C_t),$$

subject to the flow budget constraint $P_t C_t + Q_t B_t + P_t T_t = P_t Y + B_{t-1}$, where $\beta < 1$ is the households' discount factor, P_t denotes the price of consumption goods, T_t denotes real lump-sum net taxes, and $Q_t = \frac{1}{R_{n,t}}$ is the price of the one-period government bond B_t , equal to the inverse of the gross nominal interest rate $R_{n,t}$.

The government budget constraint reads $Q_t B_t + P_t T_t = B_{t-1}$, where net taxes, T_t , coincide with the real primary surplus. The fiscal authority follows the fiscal rule:

$$\frac{\tau_t}{\tau} = \left(\frac{s_{b,t-1}}{s_b} \right)^{\gamma} e^{\zeta_t},$$

where $\tau_t = \frac{T_t}{Y}$ denotes the surplus-to-output ratio, $s_{b,t} = \frac{Q_t B_t}{P_t Y}$ denotes the real market value of debt as a share of output, τ and s_b are their respective steady-state values, ζ_t is a shock to lump-sum taxes that follows an autoregressive (AR(1)) process, and the parameter γ determines how strongly the fiscal authority adjusts primary surpluses to fluctuations in debt.

The central bank behaves according to the following monetary rule:

$$\frac{R_{n,t}}{R_n} = \left(\frac{\Pi_t}{\Pi} \right)^{\phi},$$

where $\Pi_t = \frac{P_t}{P_{t-1}}$ is the gross inflation rate at time t , variables without the time subscript denote the corresponding steady states, and the parameter ϕ controls the strength with which the central bank reacts to movements of inflation from its target.

Combining the households' Euler equation with the market-clearing condition $C_t = Y$ in every period leads to the Fisher equation: $Q_t = \beta (\mathbb{E}_t \Pi_{t+1})^{-1}$.

2. *Linearized System of Equations.* We linearize the model equations around the deterministic steady state. Henceforth, hatted variables denote variables in log-deviation from their steady-state values. We obtain the following system of equations:

$$(1) \quad \hat{r}_{n,t} = \mathbb{E}_t \hat{\pi}_{t+1},$$

$$(2) \quad \hat{s}_{b,t} = \beta^{-1} [\hat{s}_{b,t-1} + \hat{r}_{n,t-1} - \hat{\pi}_t - (1 - \beta) \hat{\tau}_t],$$

$$(3) \quad \hat{r}_{n,t} = \phi \hat{\pi}_t,$$

$$(4) \quad \hat{\tau}_t = \gamma \hat{s}_{b,t-1} + \zeta_t.$$

Plugging the monetary rule [equation \(3\)](#) into the Fisher [equation \(1\)](#) leads to the monetary block:

$$(5) \quad \mathbb{E}_t \hat{\pi}_{t+1} = \phi \hat{\pi}_t.$$

Combining the law of motion for debt [equation \(2\)](#) with the fiscal rule [equation \(4\)](#) yields the fiscal block:

$$(6) \quad \hat{s}_{b,t} = \beta^{-1} [1 - (1 - \beta)\gamma] \hat{s}_{b,t-1} + \beta^{-1} [\hat{r}_{n,t-1} - \hat{\pi}_t - (1 - \beta)\zeta_t].$$

3. *Existence and Uniqueness of a Solution.* As shown by [Leeper \(1991\)](#), in this class of models there are two regions of the parameter space that deliver existence and uniqueness of a stationary solution. In the first region, monetary policy is active and responds more than one-to-one to deviations of inflation from its target ($\phi > 1$). The fiscal authority implements the necessary fiscal adjustments to keep debt on a stable path ($\gamma > 1$). Fiscal policy is defined as passive because it passively accommodates the behavior of the monetary authority. We label this policy combination the monetary-led policy mix. The distinctive feature of the monetary-led policy mix is that the macroeconomy is completely insulated from the fiscal block, so fiscal imbalances are irrelevant for inflation determination in equilibrium (monetary and fiscal dichotomy). This is because debt stability is achieved with fiscal adjustments.² The first panel of [Figure I](#) illustrates

2. In richer models with distortionary taxation and government purchases, fiscal variables affect the macroeconomy but through a different channel than the one analyzed here.

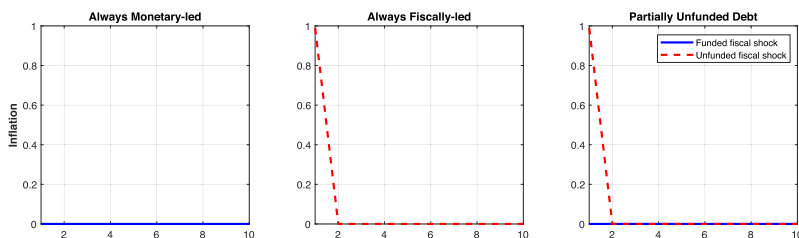


FIGURE I

Impulse Response of Inflation to a Fiscal Shock

The discount factor β is set to 0.99 and the steady-state value of the debt-to-GDP ratio s_b to 1. In the model with partially unfunded debt, the monetary policy parameters are $\phi^M = 2$ and $\phi^F = 0$ and the fiscal policy parameters are $\gamma^M = 20$ and $\gamma^F = 0$. The Always Monetary-led model is parameterized as follows: $\phi = \phi^M$ and $\gamma = \gamma^M$. The Always Fiscally led model is parameterized as follows: $\phi = \phi^F$ and $\gamma = \gamma^F$. Fiscal shocks have an autocorrelation coefficient of 0.5, and their variance is scaled to produce a unit response of inflation on impact of an unfunded shock.

this point by showing that inflation does not move in response to a negative shock to primary surpluses.

In the second region of the parameter space, labeled fiscally led policy mix, the fiscal authority is not committed to implementing the necessary fiscal adjustments. Monetary policy is now passive ($\phi \leq 1$) because it passively accommodates the behavior of the active fiscal authority ($\gamma \leq 1$). Under the fiscally led policy mix, the macroeconomy is not insulated from fiscal imbalances. In fact, inflation is determined by the need for stabilizing government debt. Consequently, fiscal imbalances affect inflation. The second panel of Figure I illustrates this point. Now a negative shock to primary surpluses leads to an increase in inflation. This increase in inflation is fully accommodated by the central bank ($\phi = 0$), and debt stability is preserved.

4. Shock-Specific Rules and Partially Unfunded Debt. We extend the model to allow the monetary-led and fiscally led policy mixes to coexist. In this new class of models, the dynamics typical of a monetary-led policy mix coexist with the dynamics typical of a fiscally led policy mix. We focus on fiscal shocks, but the logic applies to all types of shocks that move the fiscal burden of the economy, as illustrated in the richer model considered in our empirical analysis. In what follows, we use superscripts M and F to denote policy parameters that imply a behavior in line with a monetary-led policy mix and a fiscally led policy mix, respectively.

We consider the following fiscal rule:

$$(7) \quad \frac{\tau_t}{\tau} = \left(\frac{s_{b,t-1}}{s_{b,t-1}^F} \right)^{\gamma^M} \left(\frac{s_{b,t-1}^F}{s} \right)^{\gamma^F} e^{\zeta_t^M + \zeta_t^F},$$

where ζ_t^M and ζ_t^F denote funded and unfunded fiscal shocks, respectively. With respect to the amount of unfunded debt $s_{b,t}^F$ accumulated as a result of the unfunded fiscal shocks, the fiscal authority is not committed to implementing a large enough fiscal adjustment: $\gamma^F \leq 1$. Instead, the fiscal authority is willing to fully stabilize deviations of debt from its unfunded component: $\gamma^M > 1$. Thus, fiscal policy is passive with respect to the funded component of debt ($\frac{s_{b,t-1}}{s_{b,t-1}^F}$), whereas it is active with respect to the unfunded component of debt ($s_{b,t-1}^F$).

The new monetary rule is:

$$(8) \quad \frac{R_{n,t}}{R_n} = \left(\frac{\Pi_t}{\Pi_t^F} \right)^{\phi^M} \left(\frac{\Pi_t^F}{\Pi} \right)^{\phi^F},$$

where Π_t^F denotes fiscal inflation, that is, the amount of inflation that is tolerated by the central bank due to unfunded fiscal shocks. With respect to fiscal inflation, monetary policy is passive: the central bank reacts less than one-to-one, $\phi^F \leq 1$. Instead, the central bank is active in stabilizing inflation in deviations from fiscal inflation: $\phi^M > 1$.

Linearizing the fiscal rule in [equation \(7\)](#), we obtain:

$$(9) \quad \hat{\tau}_t = \gamma^M (\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F) + \gamma^F \hat{s}_{b,t-1}^F + \zeta_t^M + \zeta_t^F.$$

Given that $\gamma^F \leq 1$, the expression makes clear that the fiscal adjustments are not large enough to cover the entirety of the fiscal burden.

Linearizing the monetary rule, we obtain:

$$(10) \quad \hat{r}_{n,t} = \phi^M (\hat{\pi}_t - \hat{\pi}_t^F) + \phi^F \hat{\pi}_t^F.$$

If we further assume $\phi^F = 0$, we obtain a Taylor rule that is isomorphic to a rule with a time-varying target: $\hat{r}_{n,t} = \phi^M (\hat{\pi}_t - \hat{\pi}_t^F)$. Crucially, the time-varying target $\hat{\pi}_t^F$ is not an additional shock; it is instead tightly related to the amount of inflation tolerated by the central bank to stabilize a portion of the overall fiscal burden, leading to a fiscal theory of persistent inflation.

[Online Appendix A](#) exploits the linearity of the model to prove that the components of debt and inflation in deviations from their

corresponding targets, $\hat{s}_{b,t}^M = \hat{s}_{b,t} - \hat{s}_{b,t}^F$ and $\hat{\pi}_t^M = \hat{\pi}_t - \hat{\pi}_t^F$, are exactly the amounts of debt and inflation that would arise if the monetary-led policy mix were always in place and only funded shocks occurred. We can interpret $\hat{s}_{b,t}^M$ as the amount of funded debt that the fiscal authority is committed to stabilize through fiscal adjustments. Analogously, $\hat{\pi}_t^M$ corresponds to movements in inflation originating from shocks that the central bank does not accommodate and that are instead the responsibility of the fiscal authority. We use the superscript M to emphasize that the monetary-led policy mix applies with respect to these variables.

Using the fact that in the linearized model the total amount of debt is the sum of two components, funded and unfunded debt, that is, $\hat{s}_{b,t} = \hat{s}_{b,t}^M + \hat{s}_{b,t}^F$, we can rewrite the fiscal rule as

$$\hat{\tau}_t = \gamma^M \hat{s}_{b,t-1}^M + \gamma^F \hat{s}_{b,t-1}^F + \zeta_t^M + \zeta_t^F.$$

Similarly, exploiting the fact that in the linearized model $\hat{\pi}_t = \hat{\pi}_t^M + \hat{\pi}_t^F$, the monetary rule can be rewritten as

$$\hat{r}_{n,t} = \phi^M \hat{\pi}_t^M + \phi^F \hat{\pi}_t^F.$$

Thus, the linearized model allows two equivalent ways to interpret the policy rules. First, the policy rules can be interpreted as describing a situation in which policy makers react to time-varying targets that are driven by the need to stabilize the amount of unfunded debt. Alternatively, the policy rules can be interpreted as shock-specific rules in which policy makers react differently to the different components of the endogenous target variables depending on the shocks that generate the fluctuations.

Substituting the monetary rule [equation \(10\)](#) into the Fisherian [equation \(1\)](#) yields the monetary block of the model with partially unfunded debt:

$$(11) \quad \mathbb{E}_t \hat{\pi}_{t+1} = \phi^M (\hat{\pi}_t - \hat{\pi}_t^F) + \phi^F \hat{\pi}_t^F.$$

Plugging the policy rules in the law of motion of debt [equation \(2\)](#) yields the fiscal block:

$$(12) \quad \hat{s}_{b,t} = \beta^{-1} [1 - (1 - \beta) \gamma^M] \hat{s}_{b,t-1} + \beta^{-1} [(1 - \beta) \hat{s}_{b,t-1}^F + \hat{r}_{n,t-1} - \hat{\pi}_t - (1 - \beta)(\zeta_t^M + \zeta_t^F)],$$

where, to simplify the exposition without loss of generality, we have assumed that the fiscal authority completely disregards the amount of unfunded debt: $\gamma^F = 0$.

To close the model, we need to characterize the dynamics of fiscal inflation, $\hat{\pi}_t^F$, and of the associated amount of unfunded

debt, $\hat{s}_{b,t}^F$. To do so, we construct a shadow economy in which the fiscally led policy mix is always in place and only the shocks to unfunded spending ζ_t^F occur. The shadow economy keeps track of fiscal inflation and the amount of unfunded debt. The monetary and fiscal blocks for the shadow economy are then:

$$(13) \quad \mathbb{E}_t \hat{\pi}_{t+1}^F = \phi^F \hat{\pi}_t^F,$$

$$(14) \quad \hat{s}_{b,t}^F = \beta^{-1} \hat{s}_{b,t-1}^F + \beta^{-1} (\hat{r}_{n,t-1}^F - \hat{\pi}_t^F) - \beta^{-1} (1 - \beta) \zeta_t^F.$$

Note that the monetary and fiscal blocks for the shadow economy are isomorphic to those in [equations \(5\) and \(6\)](#) once the parameter restrictions for the fiscally led policy mix are imposed and only unfunded shocks are allowed.

The set of [equations \(11\), \(12\), \(13\), and \(14\)](#) describe the model with partially unfunded debt. Because there are two non-predetermined variables ($\hat{\pi}_t$ and $\hat{\pi}_t^F$) and two eigenvalues outside the unit circle associated with [equations \(11\) and \(14\)](#), the model satisfies the Blanchard and Kahn conditions and is thereby determinate—there exists a unique stable rational expectations equilibrium.

The third panel of [Figure I](#) presents the impulse responses in the model with partially unfunded debt. As before, we assume that the monetary authority fully accommodates the increase in fiscal inflation: $\phi^F = 0$. In response to a funded spending shock (solid blue line; color version available online), the economy with partially unfunded debt behaves exactly as in the left panel, where policy makers always follow the monetary-led policy mix, and inflation is unaffected by the shock. In response to an unfunded spending shock, inflation increases. The economy with partially unfunded debt behaves exactly as in the middle panel, where policy makers always follow the fiscally led policy mix. The policy rules in the model with partially unfunded debt are shock-specific and policy makers respond differently depending on the nature of the fiscal shocks. Thus, the properties of the monetary-led and fiscally led policy mix coexist in the model with partially unfunded debt.

At a more intuitive level, our modeling formalism connects to a central idea of the fiscal theory of the price level, namely, that debt stability can be achieved with a mix of fiscal adjustments and movements in inflation that are accommodated by the central bank. With respect to this idea, the shadow economy serves the purpose of measuring the evolution of the share of debt that

is unfunded and expected to be covered by inflation, while the remaining share, which is funded, is expected to be covered via fiscal adjustments. In this sense, the partition between funded and unfunded spending is akin to an accounting exercise, under the typical assumption of perfect and symmetric information of both agents and policy makers.

Finally, it is worth mentioning that we could also have solved the model by constructing a different (monetary-led) shadow economy in which all public debt is funded and the central bank always follows the Taylor principle, although only funded fiscal shocks occur. This duality in solving models with shock-specific rules stems from the linearity of the model. Linearity implies that the two shadow economies are additive subeconomies of the actual economy. This means that the sum of the inflation rates and the sum of debts in the two parallel economies equal their counterparts in the actual economy (see [Online Appendix A](#)).

II.B. Unfunded Fiscal Shocks and Nominal Rigidities

We now introduce a production side to the model presented in the previous section. We use this model to investigate the role of price rigidity in the propagation of funded and unfunded fiscal shocks to inflation, output, debt, and the real interest rate.

We modify the setup by assuming the period utility function $U(C_t, N_t) = \ln C_t + \phi \ln(1 - N_t)$, where N_t represents hours worked. Households receive real wage income $W_t N_t$ in exchange for supplying labor services to the firms, and the production function is $Y_t = N_t^{1-\alpha}$. All other assumptions are the same as described in the previous section, including the specification of the fiscal rules in [equation \(7\)](#). In the case of flexible prices, which provides the benchmark for a classical economy, we assume perfect competition in both goods and labor markets. We calibrate this stylized model consistently with the parameters of the quantitative model that we estimate in [Section IV](#). The linearized equations and the calibrated parameter values are reported in [Online Appendix B](#).

The graphs in the first row of [Figure II](#) illustrate the impulse responses to funded and unfunded fiscal shocks (the blue solid and red dashed lines, respectively) in the case of flexible prices. Two points emerge from these graphs. First, expansionary fiscal shocks lead to an increase in inflation only if they are unfunded; that is, only if they are not backed by future fiscal adjustments.

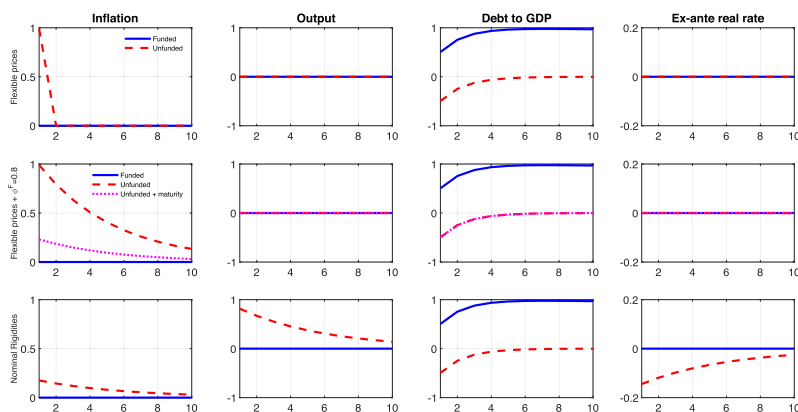


FIGURE II

Funded and Unfunded Fiscal Shocks in Production Economies

Impulse responses of inflation, real output, the debt-to-GDP ratio, and the real interest rate to funded (blue solid line) and unfunded shocks (red dashed line) to primary surpluses. The first row shows the propagation in a model with perfectly flexible prices and $\phi^F = 0$. The second row shows the propagation in a model with perfectly flexible prices and $\phi^F = 0.8$. For unfunded shocks, we also consider the case with a maturity structure (dotted magenta line). The last row reports the propagation in a prototypical New Keynesian model.

Second, regardless of whether an expansionary fiscal shock is funded or unfunded, the real economy is perfectly insulated from fiscal shocks. Like the typical model with flexible prices, the real economy is not affected by movements in inflation. Also note that the debt-to-output ratio falls in the aftermath of an unfunded fiscal shock because the increase in inflation raises nominal output (but not real output). Thus, introducing a production economy with flexible prices does not change the behavior of inflation with respect to the Fisherian model. This should not come as a surprise, because in this class of models, real activity and real interest rates are exogenous with respect to inflation.

In the second row of Figure II, we consider two extensions of the model with flexible prices. First, we allow for the central bank's response to fiscal inflation to be positive, ($\phi^F > 0$), but still less than one-to-one ($\phi^F < 1$). Without losing generality, we choose $\phi^F = 0.8$. Introducing a response to fiscal inflation larger than zero does not affect the response to funded shocks, but it determines a persistent response of inflation to unfunded fiscal shocks.

To understand why, rewrite the monetary block [equation \(13\)](#) as:

$$(15) \quad \hat{\pi}_{t+1}^F = \phi^F \hat{\pi}_t^F + \eta_{t+1}^{\pi^F},$$

where $\eta_{t+1}^{\pi^F} \equiv \hat{\pi}_{t+1}^F - \mathbb{E}_t \hat{\pi}_{t+1}^F$ is the inflation surprise in response to the unfunded fiscal shock.

The expression elucidates that the persistence of fiscal inflation is controlled by the Taylor rule coefficient ϕ^F . The central bank, by moving the nominal interest rate, sets expected inflation. If, in turn, the central bank moves the nominal interest rate in response to current inflation, the central bank induces a persistent movement in inflation. The change in expected inflation does not have a role in determining the size of the initial jump in inflation $\eta_{t+1}^{\pi^F}$. This is because anticipated inflation cannot be used to devalue one-period bonds that have not been issued yet. [Online Appendix C](#) provides additional details and shows that the initial jump only depends on the change in the present discounted value of future real surpluses.

We now make a second change and add a maturity structure for government debt while keeping $\phi^F = 0.8$. We choose an average maturity of six years. The dotted magenta line in the second row of [Figure II](#) shows how the maturity structure changes the propagation of an unfunded fiscal shock. The persistence of inflation is still controlled by the parameter ϕ^F . However, the presence of a maturity structure reduces the initial jump in inflation. This is because with a persistent movement in inflation and a maturity structure, the resulting increase in long-term interest rates devalues the outstanding long-term bonds. [Online Appendix A](#) shows analytically that the initial response declines if ϕ^F increases or if the maturity increases.

To sum up, for a given maturity, a larger ϕ^F determines a larger increase in long-term interest rates for a given inflation path, devaluing outstanding long-term bonds. For a given $\phi^F > 0$, a longer maturity implies a larger devaluation of outstanding debt. If $\phi^F = 0$ or debt is only composed of one-period bonds, the initial response collapses to what is obtained with no maturity structure. If bonds have a long maturity, but $\phi^F = 0$, inflation and short-term rates have no persistence and they cannot devalue currently outstanding long-term bonds. If $\phi^F > 0$, but debt is only short-term, there are no outstanding long-term bonds to be devalued in the first place.

Across all the cases considered so far, ex ante real interest rates and output are unaffected by the unfunded fiscal shocks.

This implies that all debt stabilization needs to be achieved via a surprise in inflation (present or future). We now make a different change to the model by introducing monopolistically competitive firms facing price rigidities. These changes make the linearized model similar to the types of stylized NK models studied in the textbooks of [Gali \(2008\)](#) and [Woodford \(2003\)](#). The linearized equations for this model are reported in [Online Appendix B](#). To isolate the effects of nominal rigidities, we revert to the $\phi^F = 0$ case and no maturity structure.

In the third row of [Figure II](#), we show the propagation of funded (blue solid lines) and unfunded shocks (red dashed line) for the model with nominal rigidities. Comparing these responses to the ones shown in the first row highlights the effects of nominal rigidities on the propagation of these two fiscal shocks. Noticeably, introducing nominal rigidities does not affect the response to funded shocks. These shocks are still irrelevant for output and inflation. However, nominal rigidities deeply affect the response to unfunded fiscal shocks. Unlike the model with fully flexible prices, after an unfunded fiscal shock, the change in inflation is spread over time, even if the model does not feature a maturity structure. Furthermore, now unfunded fiscal shocks affect the path of real interest rates and output. As a result, the increase in inflation is substantially more contained because debt stabilization is partly achieved with larger output and a prolonged decline in the cost of financing debt.

In summary, persistent movements in inflation can be obtained in a frictionless environment by allowing a positive response of interest rates to fiscal inflation. Furthermore, a maturity structure can greatly reduce the initial effect of an unfunded fiscal shock. However, movements in real interest rates and output only arise when introducing nominal rigidities. The lower cost of financing debt and the higher level of real activity contribute to stabilizing debt and mitigate the size of the initial jump in inflation.

In our quantitative evaluation, we opt for a model with nominal rigidities for several reasons. First, real interest rates and output present significant low-frequency variation in the data and play an important role in the evolution of the ratio of government debt to GDP. A model with flexible prices cannot account for these facts. Second, the lower panels of [Figure II](#) reveal some key empirical cross-correlations, which will be crucial to identify unfunded fiscal shocks in the large-scale model that we estimate

in Section IV. Namely, unfunded shocks increase inflation while decreasing both the real interest rate and the debt-to-GDP ratio. Finally, a model with flexible prices can generate persistence in fiscal inflation only if the central bank partially reacts to fiscal inflation ($\phi^F > 0$). This creates a very tight link between nominal interest rates and expected inflation that seems at odds with the data. For example, after the pandemic, inflation increased persistently even if nominal interest rates did not initially react to inflation ($\phi^F = 0$). A model with nominal rigidities in which the central bank accommodates the jump in fiscal inflation seems more appealing to account for this stylized fact.

III. A QUANTITATIVE MODEL

In this section, we build and estimate a state-of-the-art two agents NK (TANK) model with a rich fiscal block and partially unfunded debt. The model features all the ingredients that have been proven successful in matching U.S. business cycle dynamics, including a large set of business cycle shocks. With respect to these shocks, monetary policy satisfies the Taylor principle, and the fiscal authority is in control of debt stabilization. Thus, in this regard, the model behaves like its counterparts that have been extensively studied in the literature (among many others, see Christiano, Eichenbaum, and Evans 2005; Leeper, Traum, and Walker 2017). However, the model also features unfunded fiscal shocks. These shocks to transfers are not backed by future fiscal adjustments, implying that a share of the overall government debt is unfunded.³ In what follows, we outline the model in detail.

III.A. The Economy

The economy is populated by a unit measure of households, of which a fraction μ are hand-to-mouth consumers. The remaining fraction of households, $1 - \mu$, are savers, and we indicate them with an S superscript. The presence of hand-to-mouth households, together with distortionary taxation, breaks Ricardian

3. We focus on shocks to transfers because historically government purchases (G) have been constantly declining as a fraction of GDP since World War II. Thus, government purchases do not seem to represent a problem for fiscal sustainability, while transfers have been increasing over the same period. Our results are robust to allowing both types of spending to be partially unfunded.

equivalence and makes funded transfers relevant for a fraction of the population even under a monetary-led policy mix.

1. *Savers.* A household of optimizing saving agents, indexed by j , derives utility from the consumption of a composite good, $C_t^S(j)$, which comprises private consumption $C_t^S(j)$ and government consumption G_t such that $C_t^S(j) = C_t^S(j) + \alpha_G G_t$. The parameter α_G governs the substitutability between private and government consumption. When negative, the goods are complements; when positive, they are substitutes. External habits in consumption imply that utility is derived relative to the previous period value of aggregate savers' consumption of the composite good θC_{t-1}^S , where $\theta \in [0, 1]$ is the habit parameter. Saver households also derive disutility from the supply of differentiated labor services from all its members, indexed by l , $L_t^S(j) = \int_0^1 L_t^S(j, l) dl$. The period utility function is given by $U_t^S(j) = u_t^b (\ln(C_t^S(j) - \theta C_{t-1}^S) - \frac{1}{1+\xi} L_t^S(j)^{1+\xi})$, where u_t^b is a discount factor shock and ξ is the Frisch elasticity of labor supply.

Households accumulate wealth in the form of physical capital $\bar{K}_t^S$. The stock of capital depreciates at rate δ and accrues with investment I_t^S , net of adjustment costs. The law of motion for physical capital is $\bar{K}_t^S(j) = (1 - \delta) \bar{K}_{t-1}^S(j) + u_t^i [1 - s(\frac{I_t^S(j)}{\bar{K}_{t-1}^S(j)})] I_t^S(j)$, where u_t^i is a shock to the marginal efficiency of investment and s denotes an investment adjustment cost function that satisfies the properties $s(e^\kappa) = s'(e^\kappa) = 0$ and $s''(e^\kappa) \equiv s > 0$, where κ is a drift parameter capturing the log of the gross rate of technology growth in steady state.

Households derive income from renting effective capital $K_t^S(j)$ to the intermediate firms. Effective capital is related to physical capital according to $K_t^S(j) = \nu_t(j) \bar{K}_{t-1}^S(j)$, where $\nu_t(j)$ is the capital utilization rate. The cost of using one unit of physical capital is given by the function $\Psi(\nu_t(j))$. Given the steady-state utilization rate $\nu(j) = 1$, the function Ψ satisfies the following properties: $\Psi(1) = 0$, and $\frac{\Psi''(1)}{\Psi'(1)} = \frac{\psi}{1-\psi}$, where $\psi \in [0, 1)$. We denote the gross rental rate of capital as $R_{K,t}$ and the tax rate on capital rental income as $\tau_{K,t}$.

The household can also save by purchasing one-period government bonds in zero net supply and a more general portfolio of long-term government bonds in nonzero net supply. The one-period bonds promising a nominal payoff B_t at time $t + 1$ can be purchased at the present discounted value $R_{n,t}^{-1} B_t$, where the

gross nominal interest rate $R_{n,t}$ is set by the central bank. The long-term bond B_t^m mimics a portfolio of bonds with average maturity m and duration $(1 - \beta\rho)^{-1}$, where $\rho \in [0, 1]$ is a constant rate of decay. This bond can be purchased at price P_t^m , which is determined by the arbitrage condition $R_{n,t} = \mathbb{E}_t \left[\frac{1+P_{t+1}^m}{P_t^m} \right] e^{-u_t^{rp}}$, where the wedge u_t^{rp} can be interpreted as a risk premium shock.

Each period, the household receives after-tax nominal labor income, after-tax revenues from renting capital to the firms, lump-sum transfers from the government Z_t^S , and dividends from the firms D_t . These resources can be spent to consume and to invest in physical capital and bonds. Omitting the index j to simplify the notation, we can write the nominal budget constraint for the saver household as:

$$(16) \quad P_t(1 + \tau_{C,t})C_t^S + P_t I_t^S + P_t^m B_t^m + R_{n,t}^{-1} B_t \\ = \left(1 + \rho P_t^m\right) B_{t-1}^m + B_{t-1} + (1 - \tau_{L,t}) \int_0^1 W_t(l) L_t^S dl \\ + (1 - \tau_{K,t}) R_{K,t} v_t \bar{K}_{t-1}^S - \Psi(v_t) \bar{K}_{t-1}^S + P_t Z_t^S + D_t,$$

where $W_t(l)$ denotes the wage rate faced by all household members and where $\tau_{C,t}$ and $\tau_{L,t}$ denote the tax rates on consumption and labor income, respectively. The household maximizes discounted utility $\sum_{t=0}^{\infty} \beta^t U_t^S$ subject to the sequence of budget constraints in equation (16).

2. Hand-to-Mouth Households. Every period, hand-to-mouth households consume all of their disposable, after-tax income, which comprises revenues from labor supply and government transfers. The hand-to-mouth households supply differentiated labor services and set their wage to be equal to the average wage that is optimally chosen by the savers, as described below. Both savers and nonsavers face the same tax rates on consumption and labor income. Using the superscript N to indicate the nonsaving, hand-to-mouth households, we can write their budget constraint as follows:

$$(1 + \tau_{C,t}) P_t C_t^N = (1 - \tau_{L,t}) \int_0^1 W_t(l) L_t^N(l) dl + P_t Z_t^N.$$

3. Final-Good Producers. A perfectly competitive sector of final-good firms produces the homogeneous good Y_t at time t by combining a unit measure of intermediate differentiated inputs

using the technology $Y_t = (\int_0^1 Y_t(i)^{\frac{1}{1+\eta_t^p+u_t^{NKPC}}} di)^{1+\eta_t^p+u_t^{NKPC}}$, where η_t^p denotes exogenous independent and identically distributed (i.i.d.) changes in the elasticity of substitution among good varieties. In the linearized version of the model, these shocks shift the NKPC and are often dubbed cost-push shocks. The variable u_t^{NKPC} is also a cost-push shock, but it is assumed to follow a near-unit-root process. This highly persistent cost-push shock is meant to capture other external forces, such as international trade, that can generate low-frequency movements of inflation. Profit maximization yields the demand function for intermediate goods $Y_t(i) = Y_t(\frac{P_t(i)}{P_t})^{-\frac{1+\eta_t^p+u_t^{NKPC}}{\eta_t^p+u_t^{NKPC}}}$, where $P_t(i)$ is the price of the differentiated good i and where P_t is the price of the final good.

4. *Intermediate-Good Producers.* Intermediate firms produce using the technology $Y_t(i) = K_t(i)^\alpha (A_t L_t(i))^{1-\alpha} - A_t \Omega$, where Ω is a fixed cost of production that grows with the rate of labor-augmenting technological progress A_t and $\alpha \in [0, 1]$ is a parameter. The technological progress A_t follows an exogenous process that is stationary in the growth rate. Specifically, we assume that $u_t^a = (1 - \rho_a)\varkappa + \rho_a u_{t-1}^a + \varepsilon_t^a$, where $u_t^a = \ln(\frac{A_t}{A_{t-1}})$. Intermediate firms rent capital and labor in perfectly competitive factor markets. L_t is a bundle of all the differentiated labor services supplied in the economy, which are aggregated into a homogeneous input by a labor agency, as described below. Cost minimization implies that all firms incur the same nominal marginal cost $MC_t = (1 - \alpha)^{\alpha-1} \alpha^{-\alpha} (R_{K,t})^\alpha W_t^{1-\alpha} A_t^{-1+\alpha}$.

When setting prices, intermediate producers face frictions à la Calvo, that is, at time t a firm i can optimally reset its price with probability ω_p . Otherwise it adjusts the price with partial indexation to the previous period's inflation rate according to the rule $P_t(i) = (\Pi_{t-1})^{\chi_p} (\Pi)^{1-\chi_p} P_{t-1}(i)$, where $\chi_p \in [0, 1]$ is a parameter, $\Pi_{t-1} = \frac{P_{t-1}}{P_{t-2}}$, and Π denotes the aggregate rate of inflation at steady state.

Intermediate producers that are allowed to reset their price maximize the expected discounted stream of nominal profits:

$$\max \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \omega_p)^s \frac{\Lambda_{t+s}^S}{\Lambda_t^S} \left[\left(\prod_{k=1}^s \Pi_{t+k-1}^{\chi_p} \Pi^{1-\chi_p} \right) P_t(i) Y_{t+s}(i) - MC_{t+s} Y_{t+s}(i) \right],$$

subject to the demand function of the final-good sector, with Λ^S denoting the marginal utility of the savers.

5. *Wages.* We assume that both savers and hand-to-mouth households supply a unit measure of differentiated labor service, indexed by l . Each period, a saver household gets an opportunity to optimally readjust the wage rate that applies to all of its workers, $W_t(l)$, with probability ω_w . If the wage cannot be reoptimized, it will be increased at the geometric average of the steady-state rate of inflation Π and of last period inflation Π_{t-1} , according to the rule $W_t(l) = W_{t-1}(l)(\Pi_{t-1}e^{\pi})^{\chi_w}(\Pi e^{\pi})^{1-\chi_w}$, where $\chi_w \in [0, 1]$ captures the degree of nominal wage indexation. All households, including both savers and nonsavers, sell their labor service to a representative, competitive agency that transforms it into an aggregate labor input, according to the technology $L_t = (\int_0^1 L_t(l)^{\frac{1}{1+\eta_t^w}} dl)^{1+\eta_t^w}$, where η_t^w is an i.i.d. exogenous wage markup shock. The agency rents labor type $L_t(l)$ at price $W_t(l)$ and sells a homogeneous labor input to the intermediate producers at price W_t . The static profit-maximization problem yields the demand function $L_t(l) = L_t \left(\frac{W_t(l)}{W_t} \right)^{-\frac{1+\eta_t^w}{\eta_t^w}}$.

6. *Monetary and Fiscal Policy.* Assuming that one-period government bonds are in zero net supply and that households receive the same amount of transfers regardless of whether they are hand-to-mouth or savers, the government nominal budget constraint can be written as

$$(17) \quad P_t^m B_t^m + \tau_{K,t} R_{K,t} K_t + \tau_{L,t} W_t L_t + \tau_{C,t} P_t C_t \\ = (1 + \rho P_t^m) B_{t-1}^m + P_t G_t + P_t Z_t,$$

where $C_t = \mu C_t^N + (1 - \mu) C_t^S$ and $Z_t = \int_0^1 Z_t(j) dj$ denote aggregate consumption and total transfers, respectively. The budget constraint (17) implies that the fiscal authority finances government expenditures, transfers, and the rollover of expiring long-term debt by raising taxes on consumption, labor, and capital and by issuing new long-term debt obligations.

We rescale the variables entering the fiscal rules as $g_t = \frac{G_t}{A_t}$ and $z_t = \frac{Z_t}{A_t}$. For each variable x_t , $\hat{x}_t$ denotes the percentage deviation from its own steady state. Let $s_{b,t} = \frac{P_t^m B_t^m}{P_t Y_t}$ be the debt-to-GDP ratio. As in the models presented in Section II, the debt-to-GDP ratio in deviations from the steady state, $\hat{s}_{b,t}$, is the sum of two

components, funded $\hat{s}_{b,t}^M$ and unfunded $\hat{s}_{b,t}^F$ debt. We use superscripts M and F to emphasize that the monetary-led policy mix applies to funded debt, and the fiscally led policy mix applies to unfunded debt. For the shocks, we use the superscripts only to label the two types of transfers shocks, and we assume that all other shocks only affect the funded portion of debt.

The fiscal authority adjusts government spending $\hat{g}_t$, transfers $\hat{z}_t$, and tax rates on capital income, labor income, and consumption $\hat{\tau}_J$, $J \in \{K, L, C\}$ as follows:

$$(18) \quad \hat{g}_t = \rho_G \hat{g}_{t-1} - (1 - \rho_G) \gamma_G (\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F) + \zeta_{g,t},$$

$$(19) \quad \hat{z}_t^b = \rho_Z \hat{z}_{t-1}^b - (1 - \rho_Z) [\gamma_Z (\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F) + \phi_{zy} \hat{y}_t] + \zeta_{z,t},$$

$$(20) \quad \hat{\tau}_{J,t} = \rho_J \hat{\tau}_{J,t-1} + (1 - \rho_J) \gamma_J (\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F),$$

where $\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F$ denotes the portion of the debt-to-GDP ratio that the fiscal authority is committed to stabilizing with fiscal adjustments. This commitment is captured by the values for the reaction parameters γ_G , γ_Z , and $\gamma_J > 0$ that are large enough to guarantee that this portion of debt remains on a stable path. The fiscal authority does not make fiscal adjustments in response to the remaining portion of debt, which is unfunded, $\hat{s}_{b,t-1}^F$. The total amount of transfers is given by $\hat{z}_t \equiv \hat{z}_t^b + \zeta_t^M + \zeta_t^F$. The shocks ζ_t^M and ζ_t^F influence the funded and unfunded share of total transfers and are assumed to follow persistent AR(1) processes to capture the historical evolution of transfers in the United States. The term $\hat{z}_t^b$ captures transitory movements in funded transfers and possible adjustments in response to debt and the business cycle. The fiscal shocks $\zeta_{g,t}$ and $\zeta_{z,t}$ follow AR(1) processes.

The central bank moves the short-term interest rate $\hat{R}_{n,t}$ in response to fluctuations of inflation originating from the typical business cycle shocks and the funded fiscal shocks, and it fully accommodates the movements in inflation necessary to stabilize the unfunded portion of debt. As explained in Section II, this shock-specific monetary policy rule can be captured by a standard Taylor rule in which the central bank reacts to deviations of inflation from the level of inflation needed to stabilize the unfunded share of debt. We call this level of inflation tolerated by the central bank “fiscal inflation,” $\hat{\pi}_t^F$. It follows that the linearized monetary policy rule with an effective lower bound constraint

(ELB) can be written as

(21)

$$\hat{r}_{n,t} = \max \left[-\ln \underline{R}_n, \rho_r \hat{r}_{n,t-1} + (1 - \rho_r) [\phi_\pi (\hat{\pi}_t - \hat{\pi}_t^F) + \phi_y \hat{y}_t] + u_t^m \right],$$

where u_t^m is a monetary policy shock.

The parameter $\phi_\pi > 1$ implies that the Taylor principle is satisfied and monetary policy is active when responding to deviations of inflation, $\hat{\pi}_t$, from fiscal inflation, $\hat{\pi}_t^F$. The variable $\hat{\pi}_t^F$ measures the increase in inflation, relative to the central bank's long-term target (and steady-state rate), which the central bank accommodates so as to stabilize the amount of unfunded debt $\hat{s}_{b,t-1}^F$. The policy mix characterized by equations (18)–(21) implies that monetary policy is active in response to deviations of inflation from fiscal inflation and passive (no response) with respect to the inflation needed to stabilize the share of unfunded debt in deviations from its long-term target. Concurrently, fiscal policy is passive with respect to its commitment in stabilizing the share of funded government debt $\hat{s}_{b,t-1} - \hat{s}_{b,t-1}^F$, and active (no response) with respect to the unfunded share of debt. Thus, a monetary-led policy mix with respect to the typical business cycle shocks coexists with a fiscally led policy mix with respect to the unfunded fiscal shocks.

The way fiscal inflation $\hat{\pi}_t^F$ enters the Taylor rule is similar to a time-varying target or an inflation drift that is typically added to estimated medium-scale dynamic stochastic general equilibrium (DSGE) models to explain persistent inflation in the data (e.g., Smets and Wouters 2007). However, while the inflation drift in these other models evolves exogenously according to a close-to-random-walk process, fiscal inflation in our model varies in response to the need to stabilize the share of unfunded debt, which is endogenous. Changes in fiscal inflation $\hat{\pi}_t^F$ result from the coordination between monetary and fiscal authorities regarding the stabilization of the existing public debt.

III.B. Solving the Model

To solve the model, we first detrend the nonstationary variables to account for the unit root in the labor-augmenting technology A_t . We log-linearize the model equations around the deterministic steady-state equilibrium. The list of the log-linearized equations of the model is reported in Online Appendix D.

As in Section II, we construct a shadow economy that keeps track of the unfunded portion of debt and the associated evolution

of the endogenous variables. The shadow economy differs from the actual economy insofar as there are only unfunded fiscal shocks, and policy makers follow a fiscally led policy mix. All other model equations are identical across the actual and shadow economies. The model can be solved with standard techniques that apply to linear rational-expectations models.

IV. INFERENCE

The model is estimated using Bayesian techniques. The posterior distribution is obtained combining the priors for the model parameters with the model's likelihood function. The likelihood is evaluated with the Kalman filter.

IV.A. Data

The data set we use for estimation comprises 11 variables for the U.S. economy observed at quarterly frequency over the period 1960:Q1–2022:Q3: real per capita GDP growth, real per capita consumption growth, real per capita investment growth, a measure of the hours gap, the effective federal funds rate, the growth of average weekly earnings, price inflation based on the GDP deflator, the growth of real government transfers, the growth of government consumption and investment, the government debt-to-GDP ratio, and five-year break-even inflation. [Online Appendix E](#) shows how these series are constructed.

We treat five-year breakeven inflation as a noisy measure of inflation expectations over the next five years and include an observation error that captures variations in premia. To account for the federal funds rate being stuck at the effective lower bound for most of the 2008:Q1–2022:Q3 period, we estimate the model over two subsamples: from 1960:Q1 through 2007:Q4 and then from 2008:Q1 through 2022:Q3. When estimating the model on the latter subsample, we add to the data set the expectations for the Federal Funds rate 1 through 10 quarters ahead, based on overnight index swaps.⁴ Formally modeling the lower bound for the interest rate would substantially raise the computational challenge of our empirical exercise because it would introduce a nonlinearity into the model, which requires using nonlinear Monte Carlo filters to evaluate the likelihood ([Fernández-Villaverde and](#)

4. We construct series of the market-expected federal funds rate in the same way as [Campbell et al. \(2017\)](#). See [Online Appendix E](#) for further details.

Rubio-Ramírez 2007b). We adopt a simpler approach, following Campbell et al. (2012), who use data on market-based future federal funds rates to estimate the model after 2008:Q4. Agents' expectations about future interest rates are informed by the market forecasts, which enforce the ELB in the model.

IV.B. Priors

To elicit the prior distributions for the model parameters, we follow the approach proposed by Del Negro and Schorfheide (2008). Some parameter values are fixed in the estimation or implied by steady-state restrictions. We fix the discount factor β to the value of 0.99, so that the steady-state real interest rate is broadly consistent with its sample average. The quarterly rate of capital depreciation, δ , is set to target an investment rate of 2.5%. The parameters governing the steady-state markups on wages and prices cannot be separately identified in estimation, so we set them to 0.14, following Leeper, Traum, and Walker (2017). The elasticity of output to capital in the production function α is set to the standard value of 0.33. The parameter s_{gc} , capturing the ratio of government expenditures to GDP, is set to 0.11 following Leeper, Traum, and Walker (2017). Finally, the steady-state tax rates on labor, capital, and consumption, denoted by the parameters τ_L , τ_K , and τ_C , are set to the values of 0.186, 0.218, and 0.023, respectively, also based on Leeper, Traum, and Walker (2017). The consumption tax rate τ_C is assumed to be constant, so the parameters γ_C and ρ_C are set to zero.

Figure III reports the number of outstanding government bonds of different maturities (in years). Darker areas imply that a larger number of securities of the corresponding maturity is outstanding. The green dashed line corresponds to the average maturity based on the number of bonds outstanding, and the red solid line reports the value-weighted average maturity. The first noticeable fact is that both measures of average maturity are fairly stable in the United States, fluctuating around 5.5 and 4.5 years, respectively. However, over the past 10 years, the means of the average maturities have increased to 6 and 5.8 years. Given our interest in the postpandemic increase in inflation and given that the Congressional Budget Office (2020) estimates an average maturity of six years, we set the decay rate of the maturity of long-term bonds, ρ , to 0.9593, a value that implies

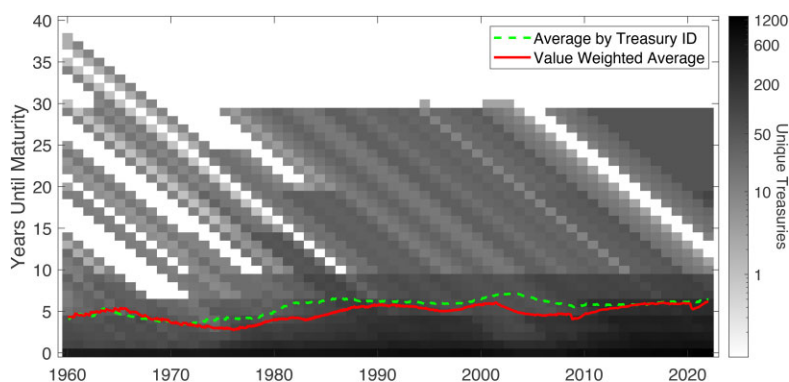


FIGURE III

Maturity on U.S. Government Debt

The figure reports the number of outstanding government bonds of different maturities (in years) over the sample 1960–2021. Darker areas imply that a larger number of securities of the corresponding maturity is outstanding. The green dashed line corresponds to the average maturity based on the number of bonds outstanding, and the red solid line reports the value-weighted average maturity.

a six-year average maturity. However, our results are robust to choosing a range of alternative values for the average maturity.

The right panels in [Tables I](#) and [II](#) report the priors for the structural parameters and for the exogenous processes, respectively. The priors for both macroeconomic and fiscal variables are generally quite diffuse. We center the prior for the share of hand-to-mouth households μ to 0.11 to match the share of poor hand-to-mouth consumers, following [Kaplan, Violante, and Weidner \(2014\)](#). The priors for the autocorrelation coefficients of the funded and unfunded transfers shocks are tightly centered around a very persistent mean to reflect the econometrician's beliefs that changes in funded and unfunded transfers are very persistent and to capture the fact that in the data transfers appear to fluctuate around a persistent trend. Conversely, cyclical increases in government transfers are expected to be backed by the increase in tax revenues and the decline in spending that ensue during the next economic recovery.

We also set the prior on the autocorrelation coefficient of the persistent cost-push shock ($\rho_{\mu NKPC}$) to provide the model with a competing mechanism to explain persistent inflation. Thus, the model allows, but does not require, persistent inflation to be generated by unfunded fiscal shocks. The autocorrelation

TABLE I
PRIORS AND POSTERIORs FOR THE EXOGENOUS PROCESSES

Param.	Description	Mode	Posterior distribution			Prior distribution		
			Median	5%	95%	Type	Mean	Std.
ρ_A^G	AR coeff. gov. cons.	0.9361	0.9372	0.9096	0.9609	B	0.500	0.100
ρ_A^U	AR coeff. funded trans.	0.9954	0.9953	0.9936	0.9967	B	0.995	0.001
ρ_A^W	AR coeff. unfunded trans.	0.9954	0.9953	0.9936	0.9967	B	0.995	0.001
ρ_A^Z	AR coeff. short-term trans.	0.4916	0.3314	0.2669	0.4590	B	0.500	0.100
ρ_a	AR coeff. technology	0.3107	0.2995	0.2156	0.3604	B	0.500	0.100
ρ_b	AR coeff. preference	0.7946	0.8033	0.7642	0.8369	B	0.500	0.100
ρ_m	AR coeff. mon. policy	0.2417	0.2613	0.2068	0.3296	B	0.500	0.100
ρ_i	AR coeff. investment	0.9218	0.9141	0.8982	0.9308	B	0.500	0.100
ρ_{rp}	AR coeff. risk premium	0.9035	0.9000	0.8844	0.9139	B	0.500	0.100
ρ_{μ^KPC}	AR coeff. pers. cost push	0.9966	0.9965	0.9953	0.9975	B	0.995	0.001
σ_G	St.dev. gov. cons.	2.0042	2.0463	1.8965	2.1828	IG	0.500	0.200
σ_M^G	St.dev. funded transfers	2.9525	2.9530	2.7788	3.2491	IG	0.500	0.200
σ_F^M	St.dev. unfunded transfers	0.5960	0.5628	0.4639	0.6674	IG	0.500	0.200
σ_Z^M	St.dev. short-term trans.	0.3897	0.3739	0.3165	0.4661	IG	0.500	0.200
σ_z	St.dev. technology	1.2159	1.2243	1.1252	1.3274	IG	0.500	0.200
σ_a	St.dev. preference	4.9930	4.9951	4.9870	4.9994	IG	0.500	0.200
σ_b	St.dev. mon. policy	0.2420	0.2446	0.2228	0.2691	IG	0.500	0.200
σ_m	St.dev. investment	0.4976	0.5007	0.4467	0.5607	IG	0.500	0.200
σ_i	St.dev. wage markup	0.3453	0.3504	0.3217	0.3864	IG	0.500	0.200
σ_w	St.dev. transitory cost push	0.1694	0.1714	0.1534	0.1920	IG	0.500	0.200
σ_p	St.dev. risk premium	0.3824	0.3994	0.3483	0.4509	IG	0.500	0.200
σ_{rp}	St.dev. persistent cost push	1.3257	1.3196	1.1878	1.6059	IG	0.500	0.200
σ_{μ^KPC}	Measur. error GDP	0.4338	0.4343	0.4001	0.4710	IG	0.500	0.200
σ_{GDP}^m	Measur. error Debt/GDP	0.3245	0.3659	0.3123	0.5153	IG	0.500	0.200
σ_{by}^m								

Notes. Posterior modes, medians, 90% posterior credible sets, and prior moments for the structural parameters. The letters in the column with the heading "Prior type" indicate the prior density function: B and IG stand for beta and inverse gamma, respectively.

TABLE II
PRIORS AND POSTERIOR FOR THE STRUCTURAL PARAMETERS

Param.	Description	Posterior distribution				Prior distribution			
		Mode	Median	5%	95%	Type	Mean	Std.	
s_b	Debt to GDP annualized	2.4582	2.4512	2.3736	2.5298	N	2.40	0.10	
100α	Steady-state growth rate	0.3979	0.3910	0.3329	0.4625	N	0.50	0.05	
$100\ln \Pi$	Steady-state inflation	0.5296	0.5333	0.4643	0.6000	N	0.50	0.05	
ξ	Inverse Frisch elasticity	1.7974	1.7440	1.6095	1.8708	N	2.00	0.25	
μ	Share of hand-to-mouth	0.0771	0.0787	0.0682	0.0906	N	0.11	0.01	
ω_w	Wage Calvo param.	0.8151	0.8167	0.7980	0.8335	B	0.50	0.10	
ω_p	Price Calvo param.	0.8673	0.8651	0.8436	0.8833	B	0.50	0.10	
ψ	Capital utilization cost	0.6564	0.6739	0.5897	0.7520	B	0.50	0.10	
s	Investment adjust. cost	5.5475	6.2053	5.4031	6.5048	N	6.00	0.50	
χ_w	Wage infl. indexation	0.0375	0.0497	0.0126	0.0824	B	0.50	0.20	
χ_p	Price infl. indexation	0.2356	0.2354	0.1908	0.3295	B	0.50	0.20	
θ	Habits in consumption	0.9134	0.9103	0.9023	0.9174	B	0.50	0.20	
α_G	Subs. private/gov. cons.	-0.0514	-0.0760	-0.1692	0.0060	N	0.00	0.10	
ϕ_y	Interest response to GDP	0.0014	0.0012	0.0001	0.0034	N	0.25	0.10	
ϕ_π	Interest response to infl.	2.0580	2.0826	1.9430	2.1966	N	2.00	0.10	
ϕ_{zy}	Transfers response to GDP	0.0823	0.0546	0.0316	0.0804	G	0.10	0.05	
γ_G	Gov. cons. response to debt	0.3443	0.3364	0.2874	0.3858	N	0.25	0.10	
γ_K	Capital tax response to debt	0.0037	0.0020	0.0003	0.0057	N	0.25	0.10	
γ_L	Labor tax response to debt	0.0027	0.0019	0.0002	0.0051	N	0.25	0.10	
γ_Z	Transfers response to debt	0.0891	0.0867	0.0359	0.1399	N	0.25	0.10	
ρ_c	AR coeff. monetary rule	0.7264	0.7284	0.6803	0.7722	B	0.50	0.10	
ρ_G	AR coeff. gov. cons. rule	0.4080	0.4139	0.3150	0.4979	B	0.50	0.10	
ρ_Z	AR coeff. transfers rule	0.5394	0.4525	0.3895	0.5843	B	0.50	0.10	

Notes. Posterior modes, medians, 90% posterior credible sets, and prior moments for the structural parameters. The letters in the column with the heading "Prior type" indicate the prior density function: N, G, IG, and B stand for normal, gamma, inverse gamma, and beta, respectively.

coefficients of the tax rules (ρ_K , ρ_L) are set to 0.5 because they are only weakly identified in the estimation. The prior for the standard deviation of the shocks is the same across the different shocks.

IV.C. Posterior Distributions

The left panels of [Tables I](#) and [II](#) report the posterior distributions for the structural parameters and the exogenous processes, respectively, obtained over the sample period 1960:Q1–2007:Q4. The estimates obtained over the second subsample, 2008:Q1–2022:Q3, are reported in [Online Appendix F](#). We note that the parameters governing the response of the tax instruments to debt (i.e., γ_L and γ_K) are positive but quantitatively small. The stabilization of the share of funded debt is therefore ensured by the relatively higher estimate of the parameters γ_G and γ_Z , implying that debt stabilization is mostly achieved by changing government spending rather than taxes. Our estimated price and wage rigidities are on the lower side relative to the literature, whereas the habit parameter lies toward the upper end of the multitude of estimates reported in the literature, but is smaller than those obtained by [Leeper, Traum, and Walker \(2017\)](#). The output coefficient in the Taylor rule is close to zero and smaller than typically obtained in the estimation of similar models, suggesting that fluctuations in the amount of fiscal inflation accommodated by the central bank dominate the interest rate response to output.

V. RESULTS

In this section, we use the estimated TANK model to assess the empirical relevance of unfunded fiscal shocks in explaining fluctuations in inflation and GDP growth. We first take a historical perspective and then focus on the pandemic period.

V.A. Identification of Unfunded Transfer Shocks

We study how unfunded transfer shocks, ζ_t^F , funded transfer shocks, ζ_t^M , and long-run cost-push shocks, u_t^{NKPC} , propagate through the economy. This analysis sheds light on how these shocks are identified in the estimation. [Figure IV](#) shows that the shocks give rise to very different impulse responses for key macroeconomic variables, that is, the inflation rate, the real interest rate, and the debt-to-GDP ratio.

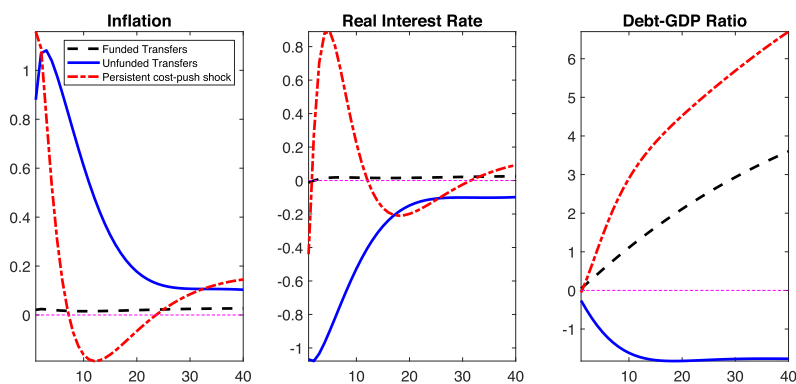


FIGURE IV

Impulse Responses for Quantitative Model

Impulse responses for inflation, the real interest rate, and the debt-to-GDP ratio to a shock to funded transfers (black dashed line), to unfunded transfers (blue solid line), and to the persistent cost-push shock (red dotted-dashed line). The units are percentage deviations from steady-state values. The magnitude of the initial shocks is set to be equal to one standard deviation as estimated in the second sample (2008:Q1–2022:Q4).

Despite the presence of hand-to-mouth households, a funded transfer shock (dashed black line) produces only a modest effect on the macroeconomy, as the expansionary impulse of current transfers is offset by the expectations of higher taxes and/or a decrease in government spending in the future; qualitatively, inflation rises, following the positive stimulus to aggregate demand and real marginal costs. Concurrently, the debt-to-GDP ratio increases to fund the rise in transfers. The increase in the debt-to-GDP ratio is very persistent because tax revenues and government purchases react very sluggishly to the increase in debt. However, debt is still on a sustainable path because the fiscal adjustments are eventually going to occur. In fact, the debt-to-GDP ratio increases exactly because it is backed by the expectation of future fiscal adjustments.

Unfunded transfers shocks, conversely, have a quantitatively strong expansionary effect on the macroeconomy (solid blue line). In sharp contrast with the propagation of funded transfers shocks, unfunded transfers lead to a fall in the real interest rate, as the monetary authority allows inflation to rise to stabilize the increase in transfers. This coordinated policy increases inflation and inflation expectations. Note that five years after the shock,

inflation remains above its long-run value. The lower real interest rate also stimulates aggregate production. Lower financing costs and higher GDP determine a fall in the debt-to-GDP ratio, despite the increase in spending. The opposite responses of the debt-to-GDP ratio produced by funded and unfunded transfers, along with the large effects on the macroeconomy from unfunded shocks that are absent in response to funded shocks, allow us to separately identify these two transfers shocks in the estimation.

A shock to firms' long-run cost-push shocks (dot-dashed red line) produces a temporary but short-lived rise in inflation. The real interest rate falls, but only on impact, and then rises persistently as the central bank reacts to the inflationary pressure. The rise in real rates and the associated contraction in aggregate production lead to an increase in the debt-to-GDP ratio. The opposite responses of the real interest rate and of the debt-to-GDP ratio following a shock to unfunded transfers and a long-run cost-push shock, provide identification for the two shocks. The response of the debt-to-GDP ratio to all three shocks is very persistent. This is consistent with the fact that the debt-to-GDP ratio is very persistent in the data. In the model, the persistence depends on the inverse of agents' discount factor (β^{-1}) and the sluggish adjustment of taxes and spending in response to funded debt.⁵

In [Online Appendix K](#), we show that other observable variables, such as GDP and hours worked, contribute to sharpening the identification of the unfunded transfer shocks as well.

V.B. Funded and Unfunded Transfers

Before using the model to infer the historical behavior of the shares of funded and unfunded transfers, it is useful to examine how total federal transfers have evolved over time. The left panel of [Figure V](#) plots the evolution of real U.S. government transfers from 1960:Q1 through 2022:Q3 (black line), in deviations from the steady state. The red dashed line corresponds to a linear trend fitted on three time periods of interest, which are marked by the vertical blue lines. The trend computed over the third period is also projected onto the pandemic subsample, 2020:Q1–

5. It takes roughly 600 quarters for the debt-to-GDP ratio to converge to steady state after an unfunded transfer shock and roughly twice as long to converge following a funded transfer shock or a long-run cost-push shock. However, debt is always on a stable path thanks to the adjustments of the other variables, which can move faster.

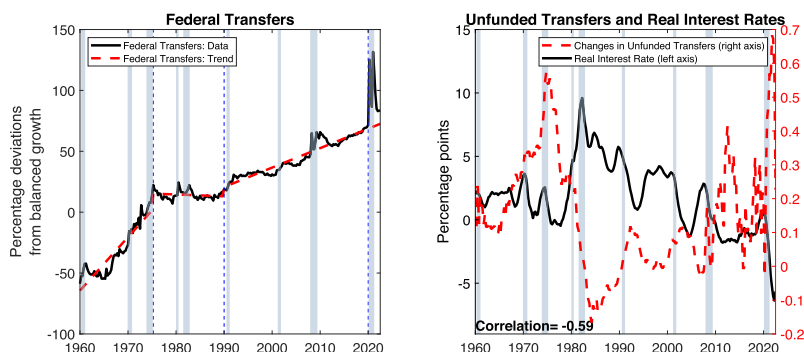


FIGURE V

Federal Transfers and Real Interest Rate

Left panel: federal transfers in percentage deviations from balanced growth (black solid line) and their linear trend fitted on each of the first three subsample periods (1960:Q1–1974:Q4; 1975:Q1–1989:Q4; 1990:Q1–2019:Q4), demarcated by the vertical blue lines. The trend computed over the third period is projected onto the last subsample, 2020:Q1–2022:Q3. Right panel: Changes in the amount of unfunded transfers (red dashed line) and ex ante real interest rate (black solid line). The former is computed by taking the one-year moving average of the quarter-over-quarter percentage changes in the amount of unfunded transfers predicted by the model (smoothed estimates). The latter is computed by taking the three-year moving average of the annualized ex ante real rate of interest predicted by the model (smoothed estimates). The sample period is 1960:Q1–2022:Q3.

2022:Q3. The first period, spanning from the mid-1960s to the mid-1970s, is characterized by a sharp increase in transfers. This increase reflects policy initiatives, initiated by President Lyndon Johnson, aimed at reducing poverty levels. These initiatives were part of the Great Society program, which aspired to reduce racial injustice and crime and to improve the environment.

After Johnson finished his second term in 1969, the level of transfers continued to increase during Richard Nixon's administration (1969–1974). This is consistent with the fact that many of the welfare programs introduced in the 1960s shifted the long-term path of spending. But in the subsequent period, which starts in the mid-1970s and ends around 1990, the growth in transfers came to a halt, and their level remained broadly unchanged. Next, after 1990, the level of transfers started to increase again; the rate of growth has been rather stable throughout this third period, although smaller than the growth rate observed between the mid-1960s and the mid-1970s. Finally, the very last period, which captures the pandemic recession, has witnessed a large

jump in the level of transfers, whose magnitude exceeds by far any increase observed over the estimation sample. After this large spike, the level of transfers remains well above the pre-pandemic trend even at the end of the sample (around 11% above).

What portion of these changes can be attributed to funded and unfunded transfers? The right plot of [Figure V](#) shows the change in the amount of unfunded transfers based on the model estimates and its relationship with the real interest rate. This plot illustrates how the structural estimation can attribute the observed changes in total transfers to the two components, funded and unfunded. The unfunded component of transfers is defined as the amount of transfers in the shadow economy where policy makers follow fiscally led policies and the only shocks hitting the economy are the unfunded transfers shocks.

The figure highlights that the real interest rate decreases when the amount of unfunded transfers increases—and vice versa. The correlation between the two series is -0.59 . This result is consistent with the impulse response functions shown in [Figure IV](#), where the real interest rate responds negatively to increases in the share of unfunded transfers. Changes in the share of unfunded transfers require monetary and fiscal coordination. Specifically, monetary policy accommodates the movements in inflation resulting from changes in unfunded transfers, leading to fluctuations in the real interest rate.

The left panel of [Figure VI](#) reports the historical evolution of unfunded transfers in percentage deviations from the balanced growth path. The display in the northeast corner of the panel zooms in on the pandemic period. The four panels on the right side of [Figure VI](#) show the contribution of changes in unfunded and funded transfers to the overall increase in transfers over the sample. These panels refer to the same four periods discussed above and highlighted in [Figure V](#). The red line corresponds to total transfers.

Changes in the amount of unfunded transfers capture revisions in expectations about the monetary and fiscal commitment to use fiscal instruments to repay the persistent flow of total transfers. For instance, total transfers may fall while the share of unfunded transfers rises (e.g., in 2020:Q4). In the estimation, the changes in the share of unfunded transfers are chiefly informed by the joint dynamics of inflation, real interest rate, and debt-to-GDP ratio, as shown in the previous subsection ([Figure IV](#)). Historical events like an exceptionally large recession, the

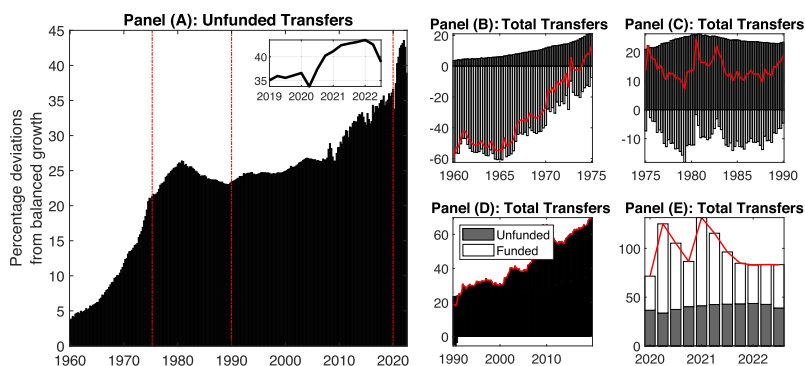


FIGURE VI

Estimated Decomposition
of Total Federal Transfers into Funded and Unfunded Components

In Panel A, we show the historical evolution of unfunded transfers in percentage deviations from the balanced growth path. The display in the northeast corner of the panel zooms into the pandemic period. In Panels B–E, we show the contribution of changes in unfunded and funded transfers to the overall increase in transfers over the sample in each of the four subsample periods (1960:Q1–1974:Q4; 1975:Q1–1989:Q4; 1990:Q1–2019:Q4; 2020:Q1–2022:Q3). The red line in these panels corresponds to total transfers. The unfunded component is the amount of transfers in the shadow economy—that is, a counterfactual economy in which the monetary and fiscal authorities follow the fiscally led policy mix and the unfunded transfers shocks (ζ_t^F) are the only exogenous disturbances. Parameters are set at their posterior mode, and shocks are estimated using the Kalman smoother. The units in all panels are percentage deviations from balanced growth. The sample is 1960:Q1–2022:Q3.

creation of large welfare programs, or the appointment of a new Federal Reserve chair can be linked to the estimated movements in the share of unfunded transfers.

The increase in transfers that occurred between the mid-1960s and the mid-1970s was partially funded, and partially unfunded (Panel B). Specifically, the rise in the amount of unfunded transfers over this period is substantial. Although the overall amount of transfers remains fairly stable between the mid-1970s and 1990, the amount of unfunded transfers exhibits a strong hump-shaped pattern (Panel C). The amount of unfunded transfers keeps increasing in the second half of the 1970s, but at a lower rate relative to that of the first half. It then starts declining rapidly in the early 1980s and by the end of this subsample, it reaches levels seen in the mid-1970s.

As we discuss in more detail later, the initial acceleration in the rise of unfunded transfers in the late 1960s, the subsequent

slowdown in the second half of the 1970s, and the drop in the early 1980s all play important roles in accounting for the rise and fall in inflation. The sharp rise in the real interest rate in the first half of the 1980s—primarily due to Volcker's aggressive monetary tightening—and the concomitantly sharp rise in the debt-to-GDP ratio explain why a smaller fraction of transfers are interpreted as unfunded. As we will see, this change in the composition of spending implies a sharp reduction in the inflation rate that the central bank is expected to tolerate ($\hat{\pi}_t^F$). Concurrently, the overall level of transfers (red line) exhibits quite an erratic behavior, which mainly affects the funded share of transfers (the white bars). These movements are mostly due to a volatile economy, with two large recessions in the late 1970s and early 1980s.

Panel D illustrates that the steady rise in transfers observed after 1990 was partially unfunded; the black bars rise steadily, albeit sluggishly, over this period. In the postmillennial period, the model recovers a more rapid increase in the share of unfunded debt observed in the 2010s in light of a very accommodating monetary policy that engendered a decade-long negative real interest rate (Bianchi et al. 2021). As shown in the right panel of Figure V, this pattern corresponds to a decline in the real interest rate, which is interpreted by the model as a sign that the central bank is willing to tolerate a higher amount of inflation.

Figure VI, Panel E shows that the federal government increased total transfers sharply in the second quarter of 2020, in an attempt to combat the severe consequences of the pandemic crisis, and then again in the first quarter of 2021. The amount of unfunded transfers slightly fell in the second quarter of 2020 and only increased in the third and fourth quarters of 2020. Interestingly, the increase in the expected share of unfunded transfers happened concomitantly with the introduction of the new monetary framework in the last days of the third quarter of 2020 (August 27). The new framework contemplates the possibility for the Federal Reserve to let inflation overshoot its 2% target after the pandemic recession. This new monetary policy strategy is reflected in the change of the expected path of the future Federal Funds rate, which we observe in the estimation using overnight index swaps. Finally, the ARPA fiscal stimulus determined a further increase in the share of unfunded transfers. As we will see in the following sections, this last policy intervention played a key role in driving the postpandemic surge in inflation.

V.C. Drivers of Inflation and GDP growth

We turn our attention to the relationship between unfunded transfers and the historical dynamics of inflation and GDP growth. [Figure VII](#) provides a historical shock decomposition of inflation and GDP growth. The black bars illustrate the evolution of inflation and GDP growth originating from unfunded transfer shocks. The gray and white bars highlight the role of the other policy shocks and the nonpolicy shocks, respectively.⁶

The key result that emerges from [Figure VII](#) is that fiscal inflation—the amount of inflation due to unfunded transfers—accounts for the bulk of inflation dynamics. The persistent rise in inflation between the mid-1960s and the mid-1970s is largely explained by the inflationary effects of the rise in unfunded transfers that took place in that period, as illustrated in [Figure VI](#). This increase in fiscal inflation, accommodated by the central bank, contributes to supporting growth in the 1970s, counteracting the productivity slowdown of those years.

The inflationary effects of this rise in the level of transfers started to wither away during the late 1970s, while nonpolicy shocks were pushing up on the rate of inflation. Even though the share of unfunded transfers rose in the second half of the 1970s ([Figure VI](#), Panel C), the pace of this increase was not fast enough to sustain the high level of fiscal inflation caused by the large expansion in unfunded transfers of the first half of the 1970s ([Figure VI](#), Panel B). As a result, fiscal inflation fell steadily in the second half of the 1970s, even if it remained elevated at the end of the 1970s, when unfunded transfers still explain about half of observed inflation. The decline in fiscal inflation contributes to the decline in growth of the late 1970s.

The fall in inflation and GDP growth accelerated in the first half of the 1980s—mostly driven by a fall in fiscal inflation. In the first five years of the 1980s, fiscal inflation declined by 3%, moving from 3.8% to 0.8% in deviations from the steady state. The sharp increase in the real interest rate due to the aggressive monetary tightening conducted by the Federal Reserve under Volcker coincides with a large fall in fiscal inflation and output growth, in the first five years of the 1980s. Thus, as in [Sims and Zha \(2006\)](#), we document that a policy change occurred before the

6. Other policy shocks include the monetary policy shock, funded transfers shocks, and the other fiscal shocks. A detailed historical decomposition of the role played by each one of these policy shocks is shown in [Online Appendix Figure G.1](#). Nonpolicy shocks include all other shocks.

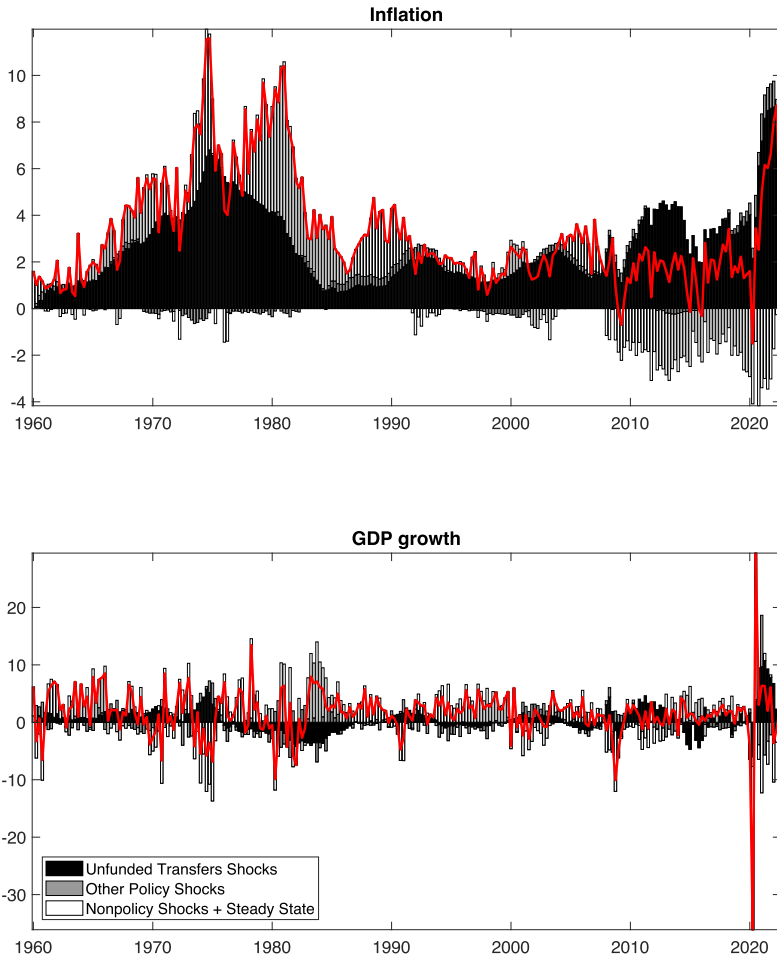


FIGURE VII

Drivers of Inflation and GDP Growth

The red solid lines correspond to annualized quarterly inflation (top panel) and annualized GDP growth (lower panel). The bars represent the cumulative contributions of unfunded transfers shocks (black bars), other policy shocks (gray bars), and nonpolicy shocks (white bars) on the two variables. The white bars also include the steady state and the initial conditions for the two variables and, for GDP growth, the measurement error. Other policy shocks include shocks to funded transfers, shocks to government purchases, and unanticipated and anticipated monetary policy shocks. Shocks are estimated using the Kalman smoother with the model parameters set at their posterior mode.

appointment of Volcker in August 1979. However, the early 1980s led to an acceleration in the change of the policy environment, propelled by the election of President Ronald Reagan, who arguably provided the political backing to the resolute disinflation policy of the Federal Reserve (Samuelson 2008; Bianchi and Ilut 2017). Thus, the rapid decline in inflation, the slowdown in real activity, and the increase in real rates of the early 1980s are interpreted as a joint monetary-fiscal policy phenomenon.⁷ These results are also consistent with the evidence provided by Hazell et al. (2022). They argue that the Phillips curve has always been flat in the United States and that the primary force behind the Volcker disinflation was a change in long-term inflation expectations triggered by the policy change, rather than high unemployment working through a steep Phillips curve.

From about 1990 until the most recent years, fiscal inflation contributes persistently to averting deflation. After the Great Recession, fiscal inflation offsets a deflationary bias set off by nonpolicy shocks—primarily a mix of favorable investment shocks and long-run cost-push shocks. Notably, the deflationary effects of these nonpolicy forces are persistent and keep dragging inflation down for a long period of time.

Fiscal inflation increases significantly after the pandemic recession. This helps in sustaining the recovery, but at the cost of a very large persistent increase in inflation. As we will see in more detail, inflation is already on an upward trajectory following the CARES Act, but it accelerates significantly following the ARPA fiscal stimulus that was implemented with the economy already on a path to recovery. Interestingly, the first change in the amount of unfunded transfers and the associated contribution to the rebound of the economy do not coincide with the increase in fiscal transfers associated with the CARES Act, but rather with the announcement of the change of the policy strategy followed by the Federal Reserve. Thus, our model suggests that

7. Our results suggest an interesting interpretation for the finding of the seminal paper by Clarida, Gali, and Gertler (2000) that the systematic response of the Federal Reserve to inflation was weaker in the 1970s than in the following decade, a result confirmed by subsequent work by Lubik and Schorfheide (2004), Fernández-Villaverde and Rubio-Ramírez (2007a), and Bianchi (2013) in estimated DSGE models. In light of our results, the estimated coefficient of a standard Taylor rule could be interpreted as a weighted average of two different coefficients whose weights change depending on the type of shocks that determine movements in inflation.

the coordination between monetary and fiscal authorities is what triggers the large rebound of the economy. The increase in funded transfers alone has limited efficacy because it also generates an expectation of large tax increases in the future. This result holds even though we allow for hand-to-mouth consumers that immediately spend the transfers that they receive. Instead, an increase in unfunded transfers leads to a reflation of the economy, a decline in real interest rates, and an increase in real activity.

In this respect, the postpandemic dynamics are in line with the historical experience. Throughout the sample, fiscal inflation increases in the aftermath of recessions, a finding consistent with what is documented in [Hall and Sargent \(2011, 2022\)](#). The pattern is particularly visible after the 1973–1975, 1990–1991, 2001, 2007–2009, and pandemic recessions. The only noticeable exception is the recession of the early 1980s. This recession is inherently different precisely because the decline in inflation and the contraction in real activity were largely caused by a decline in the share of unfunded spending. As discussed already, the model interprets the Volcker disinflation and the associated recession as a joint monetary and fiscal phenomenon.

The historical decomposition allows us to further explain how to interpret the policy rules employed in the article. Agents and policy makers understand that spending can be covered by future fiscal adjustments or movements in inflation. At each point in time, the shadow economy serves the purpose of summarizing symmetric beliefs about the share of debt that is unfunded, making sure that movements in inflation are consistent with the cross-equations restrictions of the model. However, the underlying information flow can go either way. Agents can observe the actions of the Federal Reserve and decide how much of the stabilization will occur with inflation, or they can observe the actions of the fiscal authority and conclude that inflation must increase to keep debt on a stable path. In the background, there can also be a political-economy game between the two authorities with respect to the amount of unfunded spending. In this case, our rules would capture the final outcome of the game.

Agents and policy makers are assumed to be perfectly informed and rational, a pervasive assumption in macroeconomics that we maintain for tractability. Because of this standard—admittedly strong—assumption, the monetary and fiscal authorities as well as the private sector know precisely the share of spending that will be stabilized by fiscal inflation. As modelers,

we resort to a shadow economy to conduct an accounting exercise and track the share of unfunded spending. The assumption that policy makers respond to persistent macro variables defined in a shadow economy has a long-standing tradition in theoretical and empirical macroeconomics. A prominent example is the assumption positing that the central bank responds to the output gap, which requires the modeler to construct a counterfactual economy with flexible prices and no inefficient shocks. Even in that case, the shadow economy captures a policy-relevant economic concept identified using the structure of the model.

The coordination of monetary and fiscal policies may originate from the need to finance expensive social programs, long wars, or large fiscal stimuli. These are scenarios in which the fiscal authority might not be realistically able to raise primary surpluses, leading to inflationary pressure. An example of this situation is the launch of the Great Society initiatives, which triggered an upward shift in transfers for several years after the announcement by President Johnson in 1964. Arguably, financing these expensive long-lasting social programs with only fiscal instruments would have been politically unfeasible. This generated inflationary pressure that was met by dovish monetary policy. In the early 1980s, Volcker signaled that inflationary pressure was not going to be accommodated anymore. The Reagan administration, unlike previous administrations, refrained from interfering and advocated for a small government instead, making this policy change credible.

For a given fiscal burden, there can be situations in which the monetary authority communicates its willingness to tolerate a temporary increase in inflation that reduces the need for fiscal stabilization. Agents in the model find this increase in inflation not only credible but also necessary because they do not see any concrete action on the side of the fiscal authority to stabilize debt. This increase in inflation erodes a fraction of debt, requiring a smaller fiscal adjustment. An example of this willingness to accept a temporary increase in inflation is the revision of the Fed's framework announced in late August 2020. This announcement occurred in the aftermath of the CARES Act and can be seen as a way to trigger a rebound of the economy at a moment in which monetary policy was constrained by the zero lower bound. Against this backdrop, the ARPA generated additional inflationary pressure as the economy was already along the recovery path.

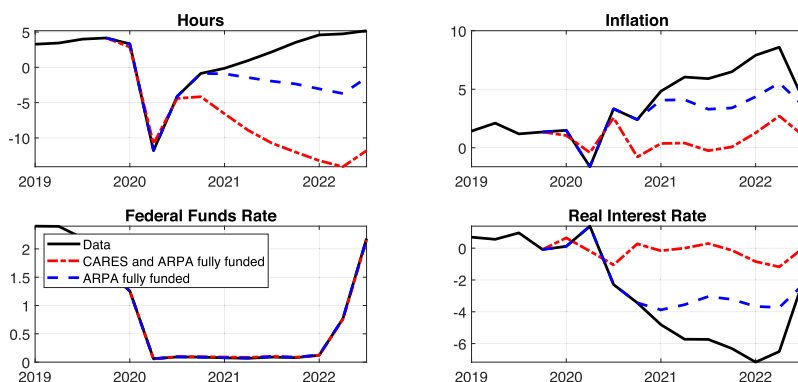


FIGURE VIII

Unfunded Fiscal Shocks and the Postpandemic Rise in Inflation

The figure compares the data (black solid line) for hours worked, inflation, the federal funds rate, and the real interest rate, with two counterfactual scenarios. In the first counterfactual simulation, all transfer shocks estimated starting from the pandemic recession (i.e., 2020:Q1) are assumed to be funded (red dashed-dotted line). In the second counterfactual simulation, only all transfer shocks estimated starting from 2021:Q1 are assumed to be funded (blue dashed line). All other shocks are left unchanged. Shocks are estimated using the Kalman smoother. Model parameters are set at their posterior mode. The estimation sample is 1960:Q1–2022:Q3.

V.D. Unfunded Fiscal Shocks during the Pandemic

We analyze in more detail the importance of monetary and fiscal policy coordination in the context of the pandemic recession. We isolate the role played by the share of unfunded fiscal transfers in boosting real economic activity and generating the increase in inflation of the postpandemic period. Figure VIII compares the realized outcomes (black solid line), with two counterfactual simulations. In the first counterfactual scenario, all transfer shocks estimated starting from the first quarter of 2021 are assumed to be funded (blue dashed line). In the second one, all transfer shocks estimated starting from the pandemic recession (i.e., 2020:Q1) are assumed to be funded (red dashed-dotted line). Thus, in the first simulation, the CARES Act and the ARPA are assumed to be fully funded, while in the second simulation only the ARPA shock is assumed to be fully funded. These two counterfactual simulations allow us to understand the relative contribution of unfunded spending associated with the two fiscal stimuli in determining the postpandemic recovery and surge in inflation.

A first key takeaway from the figure is that unfunded fiscal shocks played a central role in boosting real activity and fueling the rise in inflation. As noted earlier, this overshoot of inflation over the central bank's 2% target allows the central bank to regain space for future monetary policy in an environment of elevated ELB risk. If all transfer shocks that occurred starting from 2020 had been funded (red dotted-dashed line), the rate of inflation would have fallen in negative territory, persistently undershooting the long-run target of 2%. This deflationary bias is the consequence of the long-run drag on inflation played by a number of persistent nonpolicy shocks at the end of the sample. The highly persistent effect of these shocks on inflation is denoted by the white bars in [Figure VII](#).

Furthermore, in this alternative scenario, policy makers largely fail to rescue the economy. This can be seen by noticing that hours worked, which can be thought of as a proxy for the output gap, fall considerably in 2021. This prediction of our model should not come as a surprise given that in this alternative scenario, nominal interest rates are mired at the ELB and agents expect that the large fiscal stimuli of the pandemic period will be financed by substantial fiscal adjustments in the future. This counterfactual policy scenario compounds, instead of counteracts, the effects of adverse nonpolicy shocks (see [Figure VII](#)).

The difference between this first counterfactual (red dashed-dotted line) and the second one (blue dashed line) captures the effects of the increase in the share of unfunded transfers associated with the CARES Act in the fourth quarter of 2020. This fiscal intervention gives rise to inflationary pressure that counteracts the long-lasting deflationary forces, allowing the central bank to overshoot its long-term 2% target and avoid deflation (blue dashed line). Under this scenario, inflation is found to peak at 5.5% in 2022. The fall in real interest rates generated by the rise in inflation provides persistent stimulus to the economy, boosting total hours worked by as much as 18.8%. Furthermore, since part of the transfers paid by the government to help the economy weather the pandemic recession are expected to be unfunded, agents anticipate that future fiscal adjustments will be lighter, also contributing to the economic recovery that follows the pandemic recession.

The increase in inflation after the CARES Act stimulus is relatively modest and tolerated by the central bank, which does not anticipate the lift-off of the federal funds rate. This behavior of the nominal interest rate reflects the coordinated action of the

monetary and fiscal authorities, aimed at stabilizing a fraction of the 2020 fiscal stimulus with inflation. From this point of view, the interaction of monetary and fiscal policies plays a key role in implementing the asymmetric policy strategy outlined by the Federal Reserve after its 2020 policy review (Bianchi, Melosi, and Rottner 2021; Clarida 2020). The increase in the share of unfunded transfers makes the asymmetric strategy not only credible but also necessary.⁸

The difference between the realized outcomes (black line) and the second counterfactual simulation (blue dashed line) is due to the unfunded fiscal shocks that occurred starting from the first quarter of 2021, the quarter in which the ARPA fiscal stimulus unfolded. The ARPA fiscal stimulus helps in bringing hours worked back to the prepandemic level, counteracting contractionary nonpolicy shocks (see Figure VII). However, this large fiscal stimulus also exacerbates the preexisting inflationary pressure, arguably because it is implemented with the economy already along a recovery path. The unfunded shocks associated with the ARPA fiscal stimulus increase the peak of inflation by 3.1 percentage points, from 5.5% to 8.6%. It is worth emphasizing that this acceleration in inflation is only due to the share of unfunded spending, as the counterfactual simulations do not remove the fiscal stimuli but only assume that all spending was funded. If the ARPA had been fully funded, its inflationary effects would have been modest, despite the presence of hand-to-mouth households. This result is consistent with Parker et al. (2022), who find that the direct impact of the fiscal stimulus was modest.

So far we have conducted an in-sample analysis. We show next that our model with partially unfunded debt would have been able to predict the inflationary effects of the ARPA even out of sample. In other words, according to our model, the inflationary effects of the large fiscal stimulus were largely predictable based on where the economy was at the end of 2020 and the fact that historically fiscal expansions are partially unfunded.

V.E. Real-Time Predictions of the Effects of the ARPA

The results of the previous subsection show that inflation started increasing in response to the fiscal stimulus introduced

8. This finding echoes the following statement by Sims (2016) at the 2016 Jackson Hole symposium: “interest rate policy, tax policy, and expenditure policy, ... jointly determine the price level.”

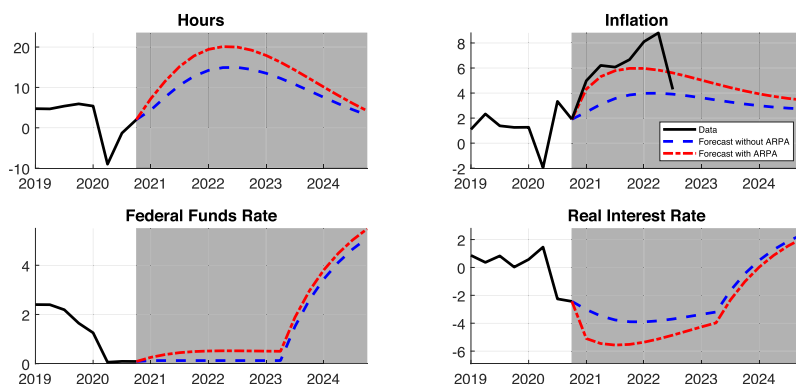


FIGURE IX

Real-Time

Out-of-Sample Projections: The Macroeconomic Effects of the ARPA Stimulus

Model forecast for hours worked, inflation, the federal funds rate, and the real interest rate, conditional on using estimates over the sample 1960:Q1–2020:Q4 and based on data available in 2021:Q2. The blue dashed line provides a real-time forecast based on filtered data through 2020:Q4 and model parameters estimated using data through 2020:Q4. The red dotted-dashed line provides a real-time forecast conditional on filtered data through 2020:Q4 and the federal transfers payment in 2021:Q1, which is when most of the ARPA stimulus checks were sent out. Shocks are estimated using the Kalman filter. Model parameters are set at their posterior mode.

by the Trump administration, but it accelerated significantly following the subsequent stimulus of the Biden administration. This additional \$1.9 trillion stimulus led influential commentators and scholars to express concern about price stability even before its implementation (e.g., [Blanchard 2021](#); [Summers 2021](#)). In this subsection, we show that our model could have predicted in real time the largely inflationary effects of the ARPA fiscal stimulus.⁹ The ability of the model to predict the postpandemic inflation is an important validation of the analysis presented in this paper.

We reestimate the model over the sample 1960:Q1–2020:Q4, only using data that were available in real time in the second quarter of 2021. We consider model projections based on real-time estimates and two alternative scenarios: with and without the ARPA fiscal shock. [Figure IX](#) reports the results. The first forecast (blue dashed line) is obtained by projecting into the future the data as of the fourth quarter of 2020. This forecast

9. In fact, the results discussed here were presented at the 2021 NBER Summer Institute.

takes into account the fiscal stimulus introduced by the Trump administration, but it excludes the ARPA stimulus enacted in the first quarter of 2021. The second forecast (red dotted-dashed line) revises this initial projection to account for the transfers observed in the first quarter of 2021, which reflect the ARPA stimulus. The data set used to produce this revised forecast differs from the data set used to produce the first forecast only to the extent that it adds information on the transfers implemented in the first quarter of 2021. In other words, we do not include observations for the other series of the model. The ARPA stimulus mostly rested on providing additional transfers to households, which received a new stimulus check in the last weeks of March 2021. Hence, the difference between the two projections isolates the contribution of the ARPA stimulus predicted by our model, everything else being equal. Importantly, the increase in transfers observed in the first quarter of 2021 is attributed by the filter to funded and unfunded shocks according to their historical patterns, using parameter estimates obtained using real-time data up through the fourth quarter of 2020.¹⁰

Figure IX shows that the ARPA stimulus produces a further increase in inflation, up from 3.7% to 6.3% at its peak. This conditional forecast can replicate very well the inflation data for 2021 (black solid line)—data points that are not used in the filtering exercise. After completing a larger overshoot, inflation retrenches to the central bank target in 2025. Because of the larger increase in inflation, the real interest rate falls further, providing an even bigger boost to the economy in 2021 and over the following years, as shown by the forecasts of hours worked (upper left panel). This is despite a modest anticipated lift-off of the nominal interest rate starting from the first quarter of 2021, relative to the baseline case. This anticipated lift-off occurs because part of the increase in transfers due to the fiscal stimulus is fiscally funded and the central bank does not accommodate the (small) increase in inflation attributed to funded fiscal shocks.

10. The decomposition of the transfers in the first quarter of 2021 into the funded and unfunded components is shown in [Online Appendix](#) Figure H.1. The amount of total transfers observed in the data as of 2021:Q2 (the real-time data set that we use in the out-of-sample forecast) is \$5.950 trillion, for a change of \$2.300 trillion between the fourth quarter of 2020 and the first quarter of 2021. The parameters of the model estimated over the shorter sample are similar to the ones estimated over the baseline sample period, shown in [Online Appendix](#) Table F.1.

The objective of this exercise is to conduct a true out-of-sample forecast using real-time data. Therefore, we control for market expectations about future monetary policy as of 2020:Q4. This is why the more aggressive lift-off in this conditional forecast only occurs in 2023, while in the data it occurs in 2022. In [Figure VIII](#), instead, we also control for the realized path of the federal funds rate and the other shocks occurring in 2021. The fact that inflation is largely predictable out-of-sample indicates that these other shocks largely affect the realized paths of GDP growth and the federal funds rate but have a relatively modest effect on realized inflation. This is consistent with the historical decomposition presented in [Figure VII](#).

This out-of-sample exercise shows that even from an *ex ante* perspective the model could foresee the rise in inflation associated with the ARPA. In other words, the model is not only able to rationalize *ex post* what happened, but also to predict the inflationary effects of a large fiscal stimulus like the ARPA. In light of these results, we conclude that the postpandemic macroeconomic outcomes are largely in line with the historical experience for the United States, where unfunded spending has played an important role in accounting for inflation dynamics. The initial fiscal interventions in response to the pandemic might have been necessary to help the recovery and move away from a low interest rate environment that limits the actions of the Federal Reserve. With the benefit of hindsight, the ARPA fiscal stimulus might have been too large, given that the economy was already on a path to recovery, and caused the acceleration in inflation starting from 2021. Without this shock, inflation would have only moderately overshoot the 2% target. As of 2023, the debt-to-GDP ratio is at a historical maximum and its service cost is on the rise. This implies that even small changes in the way debt will be stabilized can lead to large swings in inflation. An implication of these findings is that a clear path for both monetary and fiscal policies is necessary to manage inflation in a postpandemic world.

VI. FISCAL INFLATION IN A TEXTBOOK MODEL

The reason for estimating the large-scale model of [Section III](#), rather than a bare-bones NK model, is that the former features multiple competing mechanisms that can explain persistent inflation. Whether our fiscal mechanism plays a dominant role is therefore an empirical question. Indeed, in

Figure VII, we see that from the early 1990s, nonpolicy factors—primarily persistent cost-push shocks and favorable technology shocks—have persistently contributed to keeping inflation down. The model predicts, however, that fiscal inflation has played an important role in dampening these deflationary pressures due to nonpolicy factors. Estimating a simple NK model, such as the one presented in Section II.B, would not be appropriate to address the quantitative importance of our mechanism, as it rules out by construction alternative ways to account for movements in inflation. Thus, the additional features of the baseline model mostly serve the purpose of rendering the propagation of familiar business cycle shocks consistent with standard general-equilibrium models and with decades of empirical work based on vector autoregression analysis and to avoid ruling out a priori other sources of inflation fluctuations.

A drawback of estimating large models is that it is often hard to assess to what extent the results are driven by all the intricacies of the auxiliary assumptions. The purpose of this section is to show that the fiscal mechanism proposed in this article is successful in explaining persistent inflation even when it is studied using a bare-bones NK model. To make this point, we take the stylized model with nominal rigidities of Section II.B and calibrate its parameters consistently with the estimated counterparts of the quantitative NK model. We simulate the stylized model using only the shocks to unfunded transfers as estimated using the quantitative NK model. The simulated series of inflation for the simple model is shown in Figure X (black dashed line) along with the U.S. GDP deflator in the data (red dotted line). For comparison, we report fiscal inflation based on the baseline model (the blue solid line). The important takeaway is that the inflation simulated from the stylized model explains persistent inflation quite successfully.

In addition, if we compare the black dashed line with the blue solid line, which corresponds to fiscal inflation implied by the estimated NK model, we observe that the textbook NK model endowed with the fiscal mechanism proposed here generates a series of fiscal inflation that is remarkably similar to what the large NK model predicts. This is despite the fact that the shocks are not reestimated to account for the differences between the models. This indicates that what makes the quantitative model successful in accounting for the persistent movements in

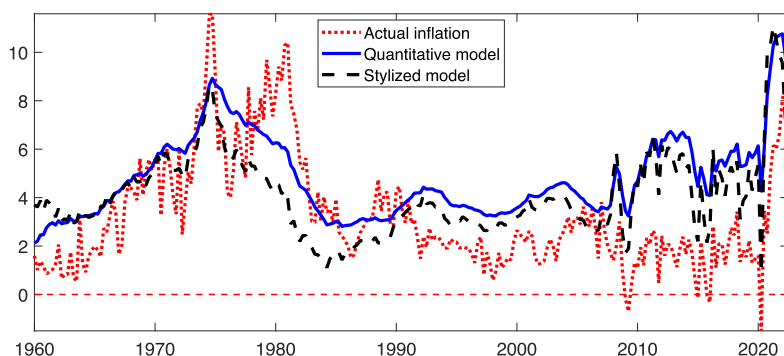


FIGURE X

Fiscal Inflation in a Textbook NK Model

The red dotted line shows the GDP deflator in the data. The blue solid line shows the inflation rate simulated from the quantitative New Keynesian DSGE model, introduced in [Section III](#), using unfunded transfers shocks. The dashed black line shows the inflation rate simulated from the stylized New Keynesian model using the same sequence of estimated unfunded transfers shocks. The sample is 1960:Q1–2022:Q3.

inflation is the presence of nominal rigidities, instead of the ancillary frictions that are popular in the NK literature.

To sum up, we conclude that the many bells and whistles of the estimated NK model do not play a key role in accounting for our results. Even when a bare-bones NK model is used in place of the large NK model, we obtain very similar results. The fiscal theory of persistent inflation arises because of the interaction between the fiscal mechanism and nominal rigidities, in line with what we explained in [Section II.B](#).

VII. CONCLUSIONS

In this article, we developed a novel class of models in which unfunded fiscal shocks coexist with typical business cycle shocks to propose a fiscal theory of persistent inflation. We first discussed the role of unfunded shocks in a series of stylized models. We incorporated unfunded fiscal shocks in a quantitative NK model. Our empirical results show that fiscal inflation plays a major role in explaining persistent movements in inflation. We use the model to study the postpandemic rise in inflation. Unfunded shocks sustained the recovery, counteracting deflationary forces. The ARPA was enacted in an economy already on a recovery path

and exacerbated the rise in inflation because it was partially unfunded. As external validation of the proposed mechanism, we show that the model is able to predict the effect of the ARPA out of sample, using real-time data. Without further unfunded fiscal shocks, inflation is expected to slowly revert to its long-term target, which is consistent with a fiscal theory of persistent inflation.

Bianchi and Melosi (2017, 2022) emphasize the role of agents' beliefs about future policy makers' behavior to explain inflation dynamics after the Great Recession and the pandemic recession, while in the current article the postpandemic rise in inflation is caused by an increase in unfunded spending. In both cases, the analysis moves away from the idea that debt is perceived as entirely backed by future taxation to make sense of the rapid increase in inflation following large fiscal expansions.

While the interpretation of the recent inflation surge provided here is qualitatively consistent with these previous contributions, there are some important differences. With respect to regime changes, the current framework allows for a continuum of intermediate outcomes between the canonical regimes and preserves the familiar propagation of shocks that do not affect the amount of unfunded debt. This captures the idea that central bankers might remain committed to long-term price stability but accommodate a temporary bout of inflation to help with fiscal sustainability. On the other hand, regime changes allow researchers to model swings in beliefs for a given policy maker's behavior, whereas in this article we assume that beliefs and policy actions are perfectly aligned. Although this is the typical assumption in most macro models, we believe that it could be relaxed in future research. In reality agents might have to learn the share of unfunded debt as policy makers could lack the ability to perfectly coordinate and communicate their policies (Melosi 2017). The speed of learning could be slow as in Bianchi and Melosi (2014), but also subject to sudden accelerations as in Bassetto and Miller (2022). Combining the approaches would allow researchers to retain the possibility of rapid swings in beliefs as in Bianchi and Melosi (2017, 2022) while preserving the desirable features of the current framework. Notably, in our estimates the share of unfunded debt typically moves sluggishly, but it is also subject to more rapid changes during important economic events such as the Volcker disinflation or the COVID-19 pandemic. This critical finding from our work suggests that the combination of the two approaches is a promising direction for future research.

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SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/VGBS6F> (Bianchi, Faccini, and Melosi 2023).

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